Welding: environment report

A robotic gas-metal-arc (MIG/MAG) welding cell in which a UR10e carries a cobot MIG gun and an FR3 with a Franka Hand handles parts at a steel fixture table, built for the "weld steel joint" action group. The robot positions the torch, tacks a bracket and lays a ¼ in horizontal fillet; the arc, heat, molten pool, bead, fusion, porosity, crater and the bond between the parts are computed, not animated, and the weld is judged the way the trade judges it (AWS D1.1 fillet gauge and macro-etched sections). This document records what was built, from which sources, with which equations and numbers, what was measured and what is not built. It does not claim an autonomous solution of any task.

Status labels used throughout: **measured** (a run of the simulation produced the number), **read** (taken from a cited source), **derived** (arithmetic from read values), **estimate** / **authored** (chosen by this build, with the reason), **open** (no digital basis; the sensitivity is reported, the value is never tuned per task).

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Every number below is from simulation runs unless marked.



Figure 0 — The shop around the cell (shop camera, rendered from the simulated fillet-2f start state, nothing commanded): the roll-up door, the windows under the eaves, high-bay pendants, the extraction canopy over the cell with the bench camera's drop rod, the welding screens on the hazard-taped cell floor, the stock and consumables shelving, the cell board with the procedure sheet and signs, the welder's disconnect, a fitter's bench and tool storage.

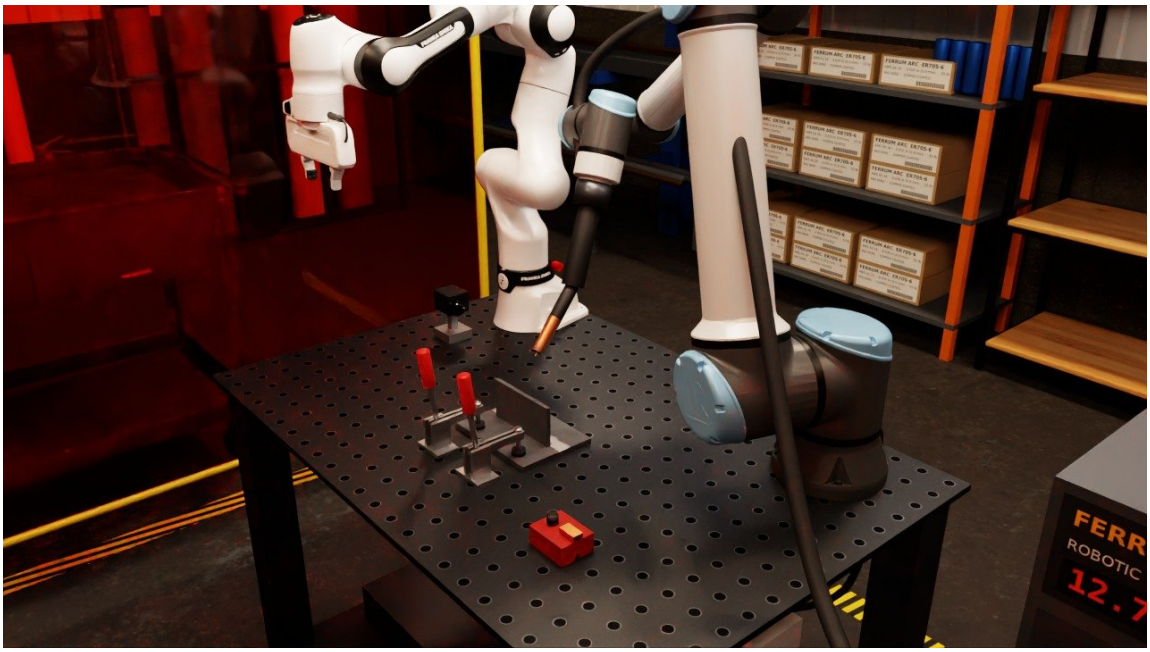


Figure 0b — The cell inside its screens (cell camera): the UR10e with the cobot MIG gun, its torch hose clipped along the arm from the wire feeder on the power source, the FR3 with the Franka Hand, the tacked coupon in its toggle clamps on the fixture table, the magnetic ground clamp and its work cable, the side camera on its magnetic base.

1. The job

The action group is 93 O*NET task rows from 52 occupations; arc and oxy-fuel fusion welding of steel is 62 % of the allocated value. The job built here is the most common robotic form of it: a cobot welding cell (UR + MIG gun packages from Vectis, Hirebotics, Lorch; FANUC CRX and Yaskawa HC10 welding cobots). A fitter clamps the parts on a steel fixture table, tacks them, and the robot runs a fillet along the joint with a constant-voltage power source whose start, gas, wire-feed and voltage the robot controller commands over its I/O (Fronius TPS/i, Kemppi AX and Miller Auto-Continuum interface signals, read). The flagship procedure is Lincoln's published horizontal-fillet spray schedule for 0.035 in ER70S-6 with 90 % Ar / 10 % CO_2: 500 in/min (12.7 m/min) wire feed, 29 V, ¾ in contact-tip-to-work, about 230 A (Lincoln C4.200, read).

What makes it hard for a robot (and a welder): the bead's size, shape and fusion are set together by travel speed, torch aim, work and travel angles and stand-off; the constant-voltage arc changes its current by about 6 A for every millimetre of stand-off; shielding is lost at a long stand-off, a steep travel angle or in a draft; the start of a weld is cold and the end leaves a crater that must be filled; a tack must fuse both members to hold.

2. Scene at real scale

Physical geometry (`welding/layout.py`, `welding/scene.py`; MuJoCo via Newton 1.5):

Item	Model	Basis
Fixture table	1200 × 800 mm steel top, 12 mm thick, 850 mm high, on four 80 mm legs	Siegmund System 16 class, read
Torch arm	UR10e (URDF of Universal Robots' ROS 2 description, pinned commit 6662e15) on the back-right corner	authored: arms on table corners as cobot carts mount them
MIG gun	2.76 kg cobot gun on the UR10e flange: mount, body, 22° neck, tapered ⅝ in bore nozzle (27 → 19 mm outside; one convex cone in collision), contact tip recessed 3.2 mm, TCP 349 mm	Tregaskiss BA1 / CA3, read; geometry authored (`welding/torch.py`)
Handling arm	FR3 + Franka Hand on the back-left corner	shared catalog (`robots/`)
Coupon	A36 base plate 200 × 150 × 10 mm (2.36 kg) and bracket 150 × 75 × 8 mm (0.71 kg), free steel bodies	authored
Hold-down clamps	two DESTACO 225-U class clamps; their spindle pads on vertical slides preloaded to 400 N each on the plate	DESTACO, read (2.2 kN capacity); preload authored
Ground (work) clamp	500 A magnetic clamp, 1.3 kg, 490 N pull toward steel within 1 mm	Strong Hand GM205 class, read
Power source	691 × 444 × 717 mm case (H × W × D) on a running gear (deck at 160 mm), front panel toward the aisle; static box collider	Miller Auto-Continuum 350, read from the maker's spec sheet AU/10.0 (Mar. 2022, p. 8: H 27.19 × W 17.5 × D 28.22 in, 130 lb; Miller's web page prints 27.2 × 17.8 × 27.9 in, 127 lb, as for the Continuum 350); deck height estimate
Wire feeder	351 × 414 mm case on the source's top at the back, 600 mm long; static box collider; the torch hose leaves its front	Continuum class feeder, read (H 13.81 × W 16.31 in); length not published: estimate
Shielding gas cylinder	Ø 235 × 1397 mm, chained in the gear's rack behind the source; static cylinder collider	size-300 cylinder, read (9.25 × 55 in)
Camera mounts	the bench camera on a 30 mm drop rod from a strut across the extraction canopy (z 2.8 m), straight above the joint, the side camera on a magnetic base on the table; 60 mm camera bodies just behind the lenses; static colliders	authored
Welding screens	three 1.83 × 1.83 m orange vinyl curtains on frames, 0.25 m off the floor; static colliders 20 mm thick	EN ISO 25980 class, 6 × 6 ft frames

Contacts are MuJoCo's soft contacts with a 4 ms time constant (the stiffest stable at the 1 ms step); at MuJoCo's default 20 ms the clamp pads sink 8 mm into the plate. MuJoCo's midphase is switched off (its per-body bounding trees, built at compile time, do not cover geoms Newton moves afterwards, so contacts with the Franka fingertips, the FR3 and UR10e links and the torch nozzle are silently dropped with it on; E19), and its no-slip post-solver (all 30 iterations every step: MuJoCo's default tolerance stops it after 4) makes static friction hold without creep. Like every other shape in the cell, the G1's shapes generate no inactive far contacts.

Everything drawn within an arm's reach collides: the equipment, the camera mounts and the screens are static colliders built from the same numbers the renderer draws (`layout.source_boxes`, `CAMERA_MOUNTS`, `screen_boxes`). Every deployment's start pose and the scripted torch paths are checked free of contact with every collider, and building a start state fails if any protective stop fires while it is built.

The shop (`welding/shop.py`, render-only beyond the colliders above). A 10 × 7.5 m bay of a steel-framed fabrication shop, 5.2 m to the roof deck: concrete block to 1.2 m with insulated profiled-steel cladding above, I-section portal frames and purlins, a strip of windows under the eaves (their panes emit the daylight: the sky is not modelled), a 3 × 3.6 m roll-up door with its coil housing, guides and bollards, a personnel door with a lit exit sign, painted walkway lines and hazard tape around the cell. Six high-bay pendants under the purlins (disk lights, 4500 K) and an LED panel in the extraction canopy light it. The canopy (galvanised, 1.8 × 1.4 m, its mouth at 2.6 m, a duct to the roof) stands above both arms' reach (UR10e base 0.85 m + 1.30 m reach + 0.35 m gun < 2.6 m), so it needs no collider. The welding equipment carries fictional "FERRUM ARC" panels whose meters read the procedure's 12.7 m/min and 29.0 V; the cylinder wears EN 1089-3 colours for an argon/CO_2 mixture with a content tag. Cables: the work cable from the source's negative stud to the ground clamp and the torch hose from the feeder along the torch arm (clipped at hangers on its upper arm and forearm) to the gun are drawn from the physical state each frame (render-only: the hose carries no load, §18); the gas hose, the source's primary cable to a lockable disconnect on the wall and the robot cables to the controllers on the table's lower shelf (UR e-Series control box 460 × 449 × 254 mm, read from UR's datasheet; the FR3 controller as a 19 in rack unit, estimate) are static. Around the cell: a cantilever rack of bar, tube, angle and flat stock, a consumables shelf stocked with ER70S-6 spool boxes, anti-spatter cans and bins, a cell board with the procedure sheet (`shop.wps_sheet`, written from `params.SCHEDULES`) and ANSI Z535.4-format signs (arc rays, fumes, robot cell, fire equipment), a gas cylinder cage, an oxy-fuel cart, a fitter's bench with a vise and hand tools, a drill press, a tool chest, a programming desk, a scrap skip and drums (21 pinned CC0 Poly Haven models and 6 textures, `welding/visual_assets.json` with SHA-256). The table top carries its 16 mm holes on a

50 mm grid as a texture (the physics top is flat).

The coupon's surface (`render.py surface_look`, render-only, estimates). Hot-rolled mill scale (dark blue-grey, mottled at 5 and 18 mm); a strip 25 mm either side of the joint ground bright, as a fitter preps hot-rolled plate (the physics does not distinguish scale from bare steel); on it the temper colours of the peak temperature each surface point reached in the thermal solver (straw, brown, purple, blue, grey; the tables give them for minutes at temperature, shifted here by about 50 °C for a weld's seconds); on the scale only a darkening; the bead a grey oxide with glassy silicate islands. Every coupon face is a 2.5 mm mesh whose colours are re-read from the solver's peak temperatures each frame, so the heat tint spreads beyond the bead as it does on real plate. Spatter: 2 % of the spatter volume in the arc's ledger (§5, the 5 % of melted wire not deposited) is drawn as balls of 0.15-0.5 mm stuck on the plate and bracket near the welded length (the stuck fraction is an estimate; the rest flies clear), and droplets are drawn in flight from the arc (0.2-1 mm at 1-4 m/s, cooling from 2673 K: estimates).

Process defects are drawn from the slice state. Porosity frozen into a slice is drawn as open pores covering that fraction of its surface (Delesse's principle) and of its macro sections; air under the arc darkens the bead toward black oxide with brown-grey soot beside it, on a log scale from 1 % air (TWI's onset of porosity) to the arc in air. A weld made with the gas valve closed gives a visibly black and pitted bead.

The arc's light. The arc is drawn as a core at the wire's end and a glow toward the pool whose mean luminance over a 3.3 mm field follows the current from measured blue-light radiances of GMAW arcs on mild steel (Nakashima et al. 2024: 5-118 W/cm²/sr at 100-350 A) converted with the 250 A spectrum: 2.0×10^8 cd/m² at 251 A (derived, ±30 %); the core's shape is an estimate. Its light is one light source whose units were measured in this renderer (a light's intensity is its luminance in cd/m², by a calibration of the renderer). It lights the cell: a crop of the cell camera goes from 53 to 99 grey levels when the arc strikes. The renderer's denoiser clamps pixels brighter than about 5×10^5 cd/m², so the directly seen arc and its mirror image in the pool (the pool model's own surface, Fresnel reflectance of liquid iron) are composited analytically at their true luminance; the firefly filter removes 15-20 % of the arc's indirect light in the cell camera. **Fume**: a plume rises from the arc (3.5 mg of fume per gram of wire in spray, Keane et al. 2014: 3.6 mg/s at 251 A; a Morton-Taylor-Turner point-source plume on about half the arc's convected heat, estimate; optical depth 0.45 at the arc, 0.06 at 1.5 m), drawn in the camera model as single scattering of the arc's light, because the renderer's own volume blocks the arc's light. The plume is turbulent: the mean plume times a lognormal factor of two-octave noise whose eddies grow with height, its axis meandering, laminar for the first 1.5-8 cm (its Reynolds number, derived). It leans toward the extraction canopy (a 1.0 m³/s sink, estimate). A cross draft has no direction in the physics and is not drawn. With air at the pool the rate moves toward 6.4 mg/g (globular in 100 % CO_2, Keane et al. 2014: the closest proxy found; no measurement for GMAW without gas), so with the valve shut the plume carries 6.6 mg/s at 250 A instead of the spray rate's 4.4. The render's constants and their provenance (`welding/render.py`):

Constant	Value	Provenance
arc blue-light radiance L_B	5.0 / 25 / 39 / 72 / 118 W/cm²/sr at 100-350 A	derived (Nakashima et al. 2024, abstract and Table 3)
luminance per L_B	513 cd/W	derived from the 250 A spectrum × V(λ) (±30 %)
arc luminance at 251 A	2.0×10^8 cd/m² over a 3.3 mm field	derived from the two above
arc core and glow shape	0.45-1.2 mm core, 10 % envelope	estimate (shape only)
arc root spread	1.8 mm at 251 A	read (the arc pressure's σ, Kim 2003 / Lin & Eagar)
hot steel emissivity / liquid	0.7 / 0.39	read (EN 1993-1-2; liquid at 807 nm, visible assumed equal: estimate)
oxide onset and scale	1 % air, log to the arc in air	1 %: read (TWI porosity onset); the scale: estimate
pore diameters, oxide and soot colours, liquid roughness, glare 2 %	—	estimate
fume per gram of wire	3.5 / 4.3 / 2.3 mg/g spray / globular / short circuit; 6.4 mg/g with the arc in air	read (Keane et al. 2014, Table 2); in air: estimate (Keane's globular in 100 % CO_2 as proxy)
plume turbulence, meander, canopy sink	lognormal s 0.7, eddies 1.5 plume radii, meander 0.35 radii; 1.0 m³/s	estimate (laminar length derived from the Reynolds number)
fume extinction, albedo	2000 m²/kg, 0.55	derived (Rayleigh-limit agglomerates; factor ~2)
plume entrainment, heat share	α 0.1; half of $(1 - \eta)$ V I	recalled textbook (Morton, Taylor & Turner 1956); estimate
droplets in flight	0.2-1 mm, 1-4 m/s, 60° spread	estimate
renderer's light units	intensity = luminance in cd/m²	measured (a calibration of the renderer)
HdrColor scale	0.01152 t S / N² per cd/m²	measured

Constant	Value	Provenance
weld camera	140 dB, exposed to its brightest pixel	140 dB read (Xiris datasheet); the rest estimate
temper colours and their 50 °C shift, mill scale and ground strip	as above	estimate

The standard cameras keep a measured tone curve.

Cameras. Policy: `bench` (straight above the joint, 0.94 m over the coupon, 40° field, looking the way the robots face: in the FR3 torch deployments the wrist covers about the lower third and the joint and the arc stay in view), `side` (joint height along the seam), `weld_camera` (on the gun: an HDR process camera of the Xiris XVC-1100 class, 140 dB read from its datasheet, exposed to its brightest pixel like the maker's arc photodetector). **The weld bead and the pool are drawn from the pool model's own continuous surfaces.** Each 0.5 mm slice's Young-Laplace profile is smoothed along the weld below the model's own sharing length, and the cells' surface is used only where the cells hold metal no profile describes. The drawn deposit equals the cells' metal within 0.1 %; drawing the cells' 0.25 × 0.5 mm occupancy instead would give the pool's edge a regular saw-tooth and facet the tacks. The pool is drawn liquid where the metal beneath it is at liquid fraction 0.5 or more (its outline is the thermal field's fusion and solidification fronts), glossy while the arc burns. The arc's metal-vapour core runs from the wire to the work (Hertel et al. 2009). It is composited at its measured luminance up to the first surface in front of it, with an edge where the plasma stops conducting (2.5 current spreads) instead of Gaussian tails. It lights the scene through a diffuse-only light, a small sphere inside it, and its reflections come from one specular-only cylinder light of its luminous intensity (units and split measured). Temper colours grow only in air, as the metal cools: not under the torch's gas, and as the colour of the peak reached, not a sweep through the cooler colours (which would put rainbow rings around the hot zone). The weld camera sees visible light, in which liquid steel at 1500 °C emits about 6×10^4 cd/m² against about 10^6 for the arc-lit solid and 10^8 for the column: its pool is a mirror, bright where it reflects the column (the crater's back slope). The bright, temperature-lit pools of published process-camera frames come from near-infrared monochrome cameras (Xiris XVC-750, product page read through a search summary); §18. The nozzle is a hollow tapered tube with an oxidised end. In the mid-weld still (Figure 2.2): the wire with its rounded end, the arc column a compact white cone cut where the pool's crest passes in front of it, the pool dark and glossy with bright highlights where it mirrors the column; no coloured rings (at most a faint violet line at the pool's lower edge). Not like the real thing: the pool's near surface looks folded, like crumpled foil, rather than smooth, and two thin bright vertical lines cross the right of the frame, whose source in the renderer was not found (§18). Diagnostic: `hero`, `seam`, `follow` (tracks the arc along the seam, filtered), and the films' shop cameras `robots` and `cell_left` (inside the screens), `cell_high` (over the screens' gap) and `shop` (`shop.FILM_CAMERAS`).

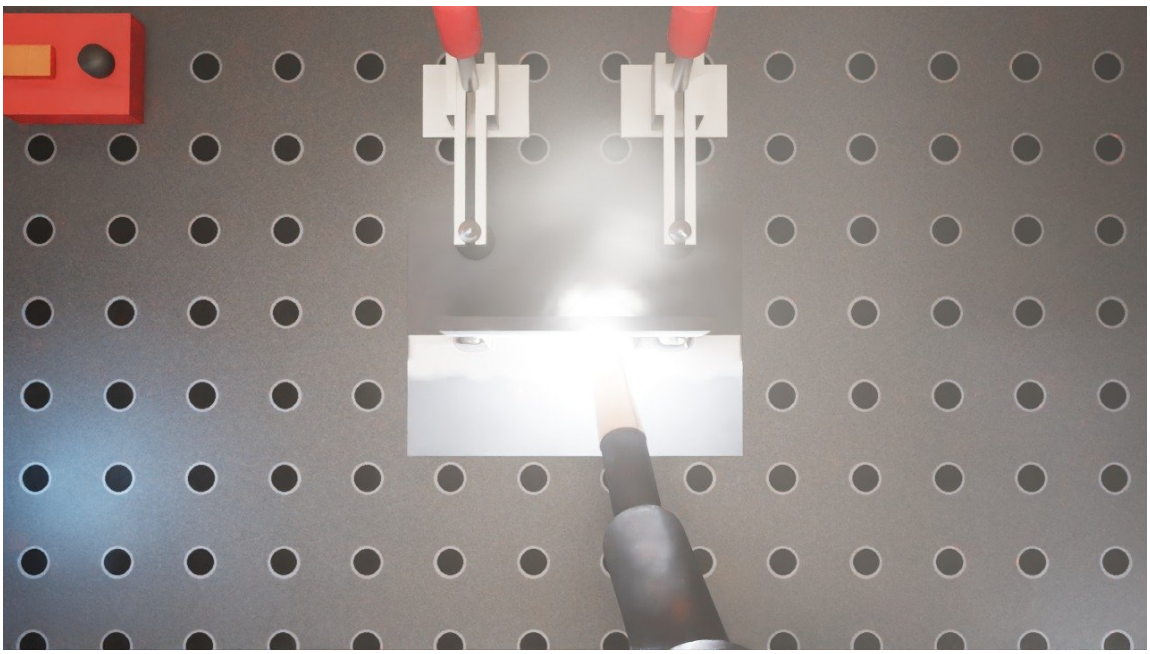


Figure 2.1 — Mid-weld, SCRIPTED ROBOT (the nominal program): the policy's bench camera, straight above the joint: the hold-down clamps (top), the base plate and the bracket's top edge, the arc at the joint, the UR10e's gun entering from below.



Figure 2.2 — The same instant through the weld camera on the gun (the HDR process camera): the arc column from the wire to the work, the pool and the bead drawn from the pool model's continuous surfaces, the nozzle's shadow at the top left.

3. The coupon in the weld frame

The process is computed in the **weld frame** carried by the base plate body: x along the joint, y out of the bracket's near (weld) face, z up from the plate; the root of the joint is the x axis and the bracket spans $x \in [0, 150]$ mm. The thermal grid (`welding/process.py` `build_grid`) covers the whole coupon: 0.5 mm along x and 0.25 mm in y and z inside the joint band $y \in [-10, 16]$ mm, $z \in [-10, 16]$ mm, growing geometrically (×1.2) to 2.5–3 mm toward the plate edges and the bracket's top: 338 × 161 × 132 = 7.18 M cells, 3.06 M of them metal.

4. Heat: conduction with melting and solidification

`welding/thermal.py` (Warp kernels on the GPU). Enthalpy finite volumes, explicit in time:

$$\frac{\partial H}{\partial t} = \nabla \cdot (k(T)\,\nabla T) + q, \qquad H(T) = \int_{T_0}^{T} \rho\, c_p\, dT' + \rho L\, f_l(H) \tag{1}$$

with f_l rising linearly from the solidus (1472 °C) to the liquidus (1512 °C) of A36 (Kim 2003, read). The latent heat is kept out of the tabulated sensible enthalpy and added analytically (`h_to_t`), so a 40 K melting range is exact on a 5 K table (a tabulated mushy range is 23 % off the Stefan solution). Steel c_p(T) and k(T) are the EN 1993-1-2 formulas (eq. 3.2, 3.3, read); the liquid's conductivity is raised to 420 W/m K, the value the Penn State conduction-only fillet models used to stand in for pool convection (Kim, Zhang & DebRoy 2003 Table II, quoted in Zhang, Kim & DebRoy 2004 part II p. 5222; read). Those models matched it to measured beads together with a 54 % arc efficiency plus 10–12 % droplet heat and a 5 mm Gaussian, on 1.32 mm wire at 287–362 A. This model's η = 0.80 puts about 1.2× that heat into the work, so the value is used outside the context it was fitted in. E3 measures that halving and doubling it move the bead's fused area by −3 % and −1 %. The same group's convective model needed 88.2 W/m K (Zhang 2004 part II Table II). Between unfused parts (bracket on plate) a contact conductance 1500 W/m² K (estimate) applies; faces in air lose h = 12 W/m² K convection plus ε σ (T⁴ − T∞⁴) radiation with ε = 0.7 (EN 1993-1-2 §2.2(2)); the plate's underside conducts into the table through the same contact conductance. Temperatures above the iron boiling point (2860 °C) are capped and the excess enthalpy is booked as evaporation. The time step is the explicit limit over the smallest cells at the largest diffusivity (0.169 ms; six substeps per 1 ms robot tick). The enthalpy is stored in float32 with each cell's float32 remainder (a compensated update): near ambient a cooling cell holds about 1.1 × 10⁸ J/m³ and changes by a few J/m³ per substep, which float32 alone rounds systematically (the energy ledger drifts 0.36 % in a 120 s soak; with the remainder it stays at 4 × 10⁻⁸, §15).

Ledger. Every joule is booked in float64: arc input, surface loss, table sink, evaporation, enthalpy carried by new weld metal cells and by cells removed; the residual Δstored − (in − out) is reported (§11, §14).

5. The arc as an electrical circuit

`welding/arc.py`. A constant-voltage source, cycle-averaged:

$$V_{set} = V_0 + R_a I + E_a L_a + I\left(R_{cable} + \rho_w L_s / A_w\right) \tag{2}$$

$$\frac{dL_s}{dt} = w_f - \left(aI + bL_s I^2\right) \tag{3}$$

$$L_a = d_{tw} - L_s \tag{4}$$

(the welding loop of a free-flight arc, linear in I; wire melting in Lesnewich's form; the arc length)

The transfer mode is not a setting. It follows from the operating point: with the commanded wire feed and voltage at the present contact-tip-to-work distance, a free-flight arc shorter than the droplet forming at the wire tip (about one wire diameter, `DIP_ARC_LENGTH`, estimate) cannot stand -- the drops bridge it and the transfer is short circuit; a free-flight arc at or above the globular-to-spray transition current, 175-185 A for 0.035 in ER70S-6 in 90/10 (180 A, Lincoln C4.200 Table 1, read), in a gas of at least 82 % argon ("the effective limit", Lincoln C4.200 p. 14, read) is spray; otherwise globular. Air drawn under the arc dilutes the argon, so with the valve closed (the arc in air) there is no spray. A small hysteresis (0.1 mm of arc, 3 A) keeps the mode from chattering at a boundary. In short circuit the arc is held at the droplet length L_dip, the stick-out is d_tw - L_dip, the current is the one that melts the wire as fast as it is fed (a I + b L_s I² = w_f) and the source's voltage sets the fraction f of the time the wire is shorted: V_load = (1 - f)(V0 + R_a I + E_a L_dip). At the free-flight arc length L_dip the two give the same point (f = 0): the current is continuous across the boundary (E2's mode map). There is no `process` tool output; a request carrying it is refused as an unknown output.

The constants: wire melting a = 3.152 × 10^{-4} m/(s·A), b = 1.859 × 10^{-4} 1/(s·A²), **derived**: a least-squares fit of the melt rate to six Lincoln chart rows (80/120/175 A short circuit at 12 mm, 195/230/275 A spray at 19 mm), each with a stick-out that agrees with its mode (CTWD - 6.4 mm for spray by Lincoln's rule, CTWD - L_dip for short circuit); the melt rates come within -9..+12 % of the rows. No published coefficient for this wire was found (Lesnewich 1958 could not be opened, and NISTIR 5037 prints "typical" values without a wire size). The arc's three constants are one published set, used unfitted: anode and cathode fall 11.55 V, arc resistance 0.03 Ω and column field 0.675 V/mm (Golob & Tovornik 2010, Table 1, read; their wire was about 1.14 mm and their gas is not stated). The stick-out's resistivity, 9.0 × 10^{-7} Ω m, is the measured resistivity of ER70S-6 wire (Guest 2014, 0.34 μΩ m cold to 1.5 μΩ m near the solidus, read from a figure) averaged over the temperature profile of a stick-out heated by its own current (derived: at the spray procedure the wire's end reaches 1270 °C to the solidus, and the mean is 0.84–0.97 μΩ m). The set of Thomsen 2005 (Table I: 0.022 Ω, 636 V/m and a 15.7 V open-arc constant, from Moore et al. 1997) is not used, because it is printed "only used for enabling numerical simulation" and a comparison against its own 15.7 V would be circular. Short circuit has no constant of its own.

d_tw is the distance from the contact tip to where the arc attaches: where the wire's axis meets metal (the arc column is stiff along the wire; the bead goes where the wire is aimed), or, if the axis misses, the nearest metal within an 8° cone. The arc exists only while the source is on, the wire reaches metal, **and that metal is connected to the ground clamp** through steel bodies in contact (MuJoCo contacts with load, the table counting as one conductor) or through the bond; it goes out when the loop current falls below 20 A, or in short circuit when no voltage is left for arcing (the wire stubs). Striking: the wire inches toward the work at 2.5 m/min and arcs on touching grounded metal; against metal with no circuit it stops (the feeder slips) and the source reports an arc start failure after 1.5 s. Touch sensing: with the output off and sensing on, the source puts 70 V on the wire and reports contact when the stick-out reaches grounded metal (Fronius, read). The wire is a rod along the torch axis: a ray from the contact tip against every collision shape except the gun's own stops it at whatever it meets first (table, clamp, a robot link, the coupon, a bead's collider), and the coupon's cells (which include the beads) decide where an arc attaches; brought closer than its stick-out, the wire is pushed back into the contact tip. Inching and the striking run-in stop at a 100 mm free stick-out (a model bound: the wire's sag is not modelled). With the contact tip closer to metal than the shortest arc (0.5 mm) on the shortest stick-out (0.2 mm) no arc ignites or burns: the tip shorts to the work.

The source's limits and the wire's balance. Wire is conserved: what melts is what was fed less the growth of the stick-out, and the ledger's `wire_balance_m3` (fed − deposited − spatter − stick-out growth) is zero to rounding in every run (E20 gates it at 0.001 mm³). Without this balance, a stick-out held at its 0.2 mm floor while the full melt rate is deposited would let 1.3 m/min at 44 V lay 9.5 times the wire fed. The source and wire have three limits:

- **burnback**: when the arc melts the wire faster than it is fed and the stick-out is gone, the arc climbs to the contact tip and fuses the wire to it. The arc goes out with a `burnback` fault that switching the output off and on does not clear, because clearing a wire welded to the tip is a maintenance job the cell cannot do. At 1.3 m/min, 44 V and 19 mm the source alone burns back 0.26 s after the strike (E2, §11.2).
- **current ceiling**: the Auto-Continuum 350 outputs 20-400 A (spec sheet AU/10.0, read). Above 400 A the model's source holds the current and its voltage falls below the set point; 25 m/min at 44 V runs at the

ceiling.

- **stubbing**: in dip transfer, a wire shorted more than 35 % of the time (estimate: no measured threshold was found; the model's own shorted fraction at Lincoln's short-circuit rows is 17-20 %), or one that needs more current than the ceiling to melt it, drives into the work. The arc goes out with a `stubbing` fault, and for 0.1 s (estimate) the drive pushes the gun back with the lesser of the stick-out's Euler buckling load $\pi^2 EI/(4L^2)$ and 150 N (estimate; no drive-roll force was found). At 12.7 m/min and 16 V that push is 48 N.

A long arc is bounded by the contact tip, not by a length of its own: raising the voltage lengthens the arc until the stick-out is gone. No sourced arc length limit was found, so none is applied. The consequence is disclosed in §18: at 12.7 m/min, 44 V and a 30 mm contact tip to work distance the model sustains a 26 mm arc at 357 A, and no source was found for the length at which a real CV arc goes out.

Heat partition. The work receives $\eta V I$ with $\eta = 0.80$: the thermal efficiency factor k of EN 1011-1 for MIG/MAG, read via TWI and SSAB, which is a factor relative to submerged arc's 1.0 ("all efficiency factors relate to the efficiency of submerged arc welding", TWI). Calorimetry gives 0.74–0.84 for GMAW (DuPont & Marder 1995, Arevalo & Vilarinho 2012), and Haelsig et al. 2014 report a spread over 15 % within one process. The droplets carry the deposited wire's enthalpy at 2673 K (Kim 2003): in spray they enter a 1 mm × 1.5 mm cylinder below the arc root (Kumar & Bhaduri's form); in short-circuit and globular transfer a surface layer one wire diameter deep under the wire (globular drops detach by gravity at low speed: estimate); the rest is a Gaussian surface flux of $\sigma = 4$ mm (open: the Penn State models used 5 mm at 312–362 A with 1.32 mm wire and a 54 % arc efficiency plus droplet heat, GTAW 1.5–3.6 mm; E3 reports its sensitivity on a bead on plate; on the fillet it is not measured) on every metal face that looks toward the arc. **Electrode efficiency** (the wire that ends in the weld; the rest is spatter and fume) by mode, Lincoln C4.200 p. 16 (read): spray 98 %, short circuit 93 %, globular 89 % (88-90 % in argon blends; the arc in air is given the same value: estimate).

6. Filler deposition and the weld pool's surface

`welding/pool.py`. Per x slice of the grid (a y-z cross-section), the bead is the liquid held by surface tension against gravity and the arc's pressure:

- **contact line**: the liquid's edge on the joint's original surface. It lies at the farthest of three places: the outermost fused points (peak liquid fraction ≥ 0.5) around the root; surface that liquid weld metal connected to the pool lies on (a tack's lap remelted beside a fillet's fused toe joins the fillet's pool, and the droplets stacked at a cold start join the first pool); and where it already was while the slice is liquid (liquid lying on steel stays there). It is kept as positions on the surface (`SliceState.contact`), reset when the slice freezes, and never moved by a chord a slice borrows from a neighbour (kept as offsets from the fused edge instead, a borrowed chord would move a tack's start 4.5 mm onto cold plate in one update, and a fillet over a tack would inherit the tack's offsets);
- **surface**: while a slice's share fits under a face of at most 88° to its chord (`POOL_TOE_SLOPE_MAX`), a height function h(u) over the chord between the contact points minimises the slice's energy per unit length

$$F[h] = \gamma \int \sqrt{1 + h'^2}\, du - \rho \int \left[(\mathbf{g}\cdot\mathbf{c})\, h + (\mathbf{g}\cdot\mathbf{n}) \frac{h^2}{2} \right] du + \int (p_{arc} - P) h \, du, \qquad h(\text{toes}) = 0,\ h \geq h_{solid} \tag{5}$$

$$\gamma \kappa = P + \rho\, \mathbf{g}\cdot\mathbf{r} - p_{arc} \tag{6}$$

(surface tension γ = 1.2 N/m, the liquid's weight, the arc pressure's work, the pressure–volume term), whose stationarity is Young–Laplace γκ = P + ρ g·r − p_arc; F is convex wherever gravity holds the liquid toward the chord, so the minimiser is unique (projected Newton with an Armijo line search and both bounds, batched over slices). The surface cannot pass through metal that is solid now, nor through a rigid body above the joint (the nozzle, a finger, a robot link: `world.World._pool_obstacles` casts rays from the chord along the slice's normal whenever a body's bounding box meets the region the surface can reach; liquid spreads beneath a body and does not lift it). Where the outermost metal at a station is metal that melted and has solidified again (weld metal, or parent metal the pool fused) the surface is pinned there: frozen metal does not flow back into the pool;

- **overhang**: a slice whose share does not fit under that face holds the rest as a convex bulge that overhangs its toes. The surface is then a polygon r(φ) about a centre on the chord's perpendicular bisector, below the face's lowest point (so an arc crater stays in view), over the angles between the two toes. It minimises γ × perimeter + ∫(ρ(−g·x) + p_arc) dA at a fixed area (`pool._PolarSurface`): volume control, which unlike a fixed pressure is stable past the half circle, where a pinned drop's pressure falls as it grows. Each radius lies between the solid the liquid rests on and the first metal beyond (the joint's surface past a toe, frozen weld metal, a body, the edge of the region the pool model may change), stations of resolidified metal are held, and the KKT system of the free radii (a tridiagonal Hessian with one border) is solved by Newton with a Levenberg shift (far from the minimum the Hessian on the area constraint's tangent space can be indefinite). The problem is not convex, so the start decides which equilibrium the solve finds: it starts from the surface the slice's liquid had at the last update, whatever its chord, so the liquid follows its own equilibrium from update to update (restarted from the height-function face after a toe moves, a strike's bulge would flip to a sheet along the cold plate and back in single 5 ms updates). Checked against the weightless 2-D drop pinned on a 4 mm chord (a contract test): the pressure is γ/R within 1 % and the area exact, at 0.3-2 times the half-disc's area (contact angles 38-118°). Why a bulge: on cold steel a liquid metal's contact freezes as it touches and its base stays pinned while liquid keeps arriving. Tavakoli & Kavehpour (ISCST 2012, read) saw "the contact base remains frozen on the target without any sign of further motion (not allowed to recede or advance) despite the continuing discharge of liquid ... while the bulk of the drop was still in a liquid state", with "instantaneous solidification ... at the moment of contact" at high Stefan number and low flow rate. Metal droplets deposited on cold same-metal substrates freeze as a "spherical pancake" with a solidification angle above 90° and cold laps under it (aluminium droplet printing, Materials & Design 2024, search result, not opened). For steel on 20 °C steel the Stefan number c(T_m − T_t)/L is about 4, outside the wax and alkane data of Schiaffino & Sonin (1997) and Herbaut et al. (2020, read), whose arrest angle grows with it. Where the bulge comes down on the metal beyond a pinned toe, the liquid rolls onto it: the contact line moves there, frozen at the contact (cold lap unless the metal under it fuses later);
- **volume**: the liquid slices share one pool pressure P, found by bisection so that the pool holds exactly the filler not already present in other slices' cells (the mass balance is closed on the cells). The excess the full slices cannot hold under their faces is held as bulges: each takes what it held past its face last update (the liquid stays where it is), the rest in proportion to what it holds, clipped to what its bounds allow. Nothing is stacked beside a toe, and nothing waits unless the pool outgrows the region the model may change (14 mm along each member). The pool's metal is all the liquid weld metal connected, in the section, to the liquid over its chord: one body of liquid is held whole, however far an overhang has spread over its own frozen skin (a pool that owned only liquid within a fixed reach would cut a strike's overhang and leave its far end standing in the air). `sections.floating_weld` finds weld metal with no face on steel; the demonstrations, E26 and E27 report it;

- **along the weld**: a liquid slice whose own surface has not fused takes a neighbour's chord only where fused slices lie on both sides within 1.5 mm (the liquid bridges a short gap); at the pool's ends the liquid's edge is the fusion front. The pool's metal is shared along x with a 1 mm kernel (surface tension along the weld) before each slice takes its own pressure;
- **arc pressure**: $F = -0.040 + 2.553 \times 10^{-4}\ I$ (N), Gaussian with $\sigma = 1.4875 + 0.001233\ I$ mm (Kim 2003 fits to Lin & Eagar, read), felt by the pool through a first-order lag of 27 ms, the capillary-inertial time $\sqrt{(\rho L^3/\gamma)}$ of 5 mm of liquid steel (without it the arc–pool loop oscillates);
- **how much a slice holds under its face**: what its own surface holds, under its weight and the arc's pressure, up to a toe slope of 88° (`pool.Pool._caps`), and never more than the pressure at which its longest free span would overhang, $2\gamma/L_{free} + \rho\, g\, L_{free}/2$;
- **where the edge advances**: onto more metal only once the slice is full, not up a face steeper than about 17° against gravity (liquid does not run uphill), at most 1 mm per update, and only where the liquid would not freeze at the contact: the effusivity-weighted contact temperature $T_c = (e_l T_l + e_s T_s)/(e_l + e_s)$, $e = \sqrt{(k\rho c)}$, must reach the solidus (`pool.Pool._contact_freezes`; the liquid's molecular 31 W/m K, 6950 kg/m³ and 808 J/kg K, the solid's EN 1993-1-2 values at its temperature). Spreading onto 20 °C plate would need liquid at 2965 °C, and onto 1200 °C metal 1715 °C, so the edge runs only onto metal the arc has already heated near melting. This is a contact-freezing estimate built for this model [authored], consistent with Tavakoli & Kavehpour's pinned base; it is not Schiaffino & Sonin's arrest-angle criterion. The edge does not draw back while the pool lives (an extend–retract cycle every update would open a one-slice gap in the bead);
- **freezing**: a slice whose liquid is gone keeps its metal. Liquid steel takes about 3 % more volume than the same metal solid (solidification shrinkage, TWI Job Knowledge 33), so the pool's slices hold 1.03 × their liquid metal. Metal that freezes keeps the shape it froze in, and the pool feeds its shrinkage, so the last liquid freezes short: the crater. A weld's frozen metal equals its deposit where its last liquid can take back the expansion the metal frozen before it still holds (the robots' fillets: within 0.02 %); a short tack and a parked pool freezing from its edges inward end slightly over it (E26's tacks up to 0.24 %; a 5 s parked arc +0.5 %, process only; E20's 15 s park, whose pool outgrew the model's region, +2.4 %; §18). The further ~7 % of thermal contraction to room temperature is not modelled: the cells hold room-temperature volumes;
- **with the arc off**: net growth of the cells in an update is the pool's own metal laid one cell differently and carries the pool's liquid enthalpy (with the arc on it is new filler at the droplets' cell enthalpy); and the cells the pool changes get temperatures consistent with their enthalpies at once (`process.update_pool`), so the next conduction step does not read a filled cell at an air cell's 20 °C. Without both, a parked pool's hottest cell would rise with the arc off;
- **cold start**: while no pool spans the root (until the arc has fused both members around it) the droplets stack where they land in a low dome around the drop point, equally in every direction, only in air they can reach from where they land; the first pool takes them in (the contact line covers the liquid lying on the surface);
- **undercut**: where the equilibrium surface lies below the original surface in melted metal, those cells become air (a groove) and their metal stays in the pool.

Cells change between air and weld metal every 5 ms; liquid that moves carries its enthalpy. Not represented: Marangoni-driven surface shape, humping, droplet-by-droplet transfer, surface ripples, the arc pressure's gradient along the pool, the droplets' momentum and the plasma's drag.

Alternatives measured and rejected: pinning every untouched station (the pool overflows in 688 of 990 updates and lays 28 mm films), a feeding cut-off (wider, lumpy beads), retraction of the edge (a one-slice gap), a weightless cap, and lumps stacked beside an arrested toe.

Weld metal is solid to other bodies. Each 2 mm of the joint has a box collider on the base plate (`scene.weld_colliders`), fitted after the pool has changed the segment's cells and none of its slices is liquid: of three orientations about the joint line (the plate's normal, the fillet face's normal, the bracket's normal), the one that covers the segment's weld cells with the least empty space; it collides with the robots, the gun, the hand and the clamps, not with the bracket and plate it lies on. The boxes' error is their empty space above the metal: over the nominal 20 mm bead of the collider check (§11.11) 0.7-1.6 mm (1 mm typical, the face's convexity and the cell steps) and 4.2 mm in the last segment, which the box covers whole while the bead tapers; 3.5-7.7 mm² of empty cross-section against the bead's ~22 mm². A torch pressed onto a bead stops on it with its nozzle 2.29 mm above the plate where the bead's cells under the nozzle's rim stand 1.9-2.25 mm; over the bare joint it reaches the plate (§11.11). The colliders' `geom_sameframe` flag is cleared at setup: compiled at the plate's centre, MuJoCo would otherwise copy the plate's pose to them every step and ignore the poses written to them.

A box is resized in one step when its segment refreezes. Under a resting body that would kick the body, because the pool lays metal into space the body holds and the box then grows around it (a refit check that rests the UR10e's nozzle on a robot-laid bead, and on the bare joint, while a phantom torch runs under it, measures 9 kick ticks without the two mechanisms below: contact forces of 74-1130 N, the nozzle thrown at up to 146 mm/s, the arm's protective stop at 240 N and 404 N). Two mechanisms prevent it: the pool's surface is bounded by bodies (above), and a refit may not reach into a body deeper than the box in use did

(`world.World._weld_box_clear`). The box's outer face is pulled in by the excess; if the box would then be thinner than a cell, the box in use is kept until the body has gone. With both (refit check: PASS): 0 kick ticks, the largest contact force 0.8 N on the bead and 2.8 N on the bare joint, the nozzle's largest speed 1.2 and 1.9 mm/s, no protective stop, the nozzle where it rested. Both mechanisms acted. The pool's surface was bounded by the nozzle at 420 776 and 249 501 station-updates. The guard kept the box in use at 42 and 89 refits and pulled its outer face in at 9 and 8, by up to 2.9 and 5.5 mm.

7. Shielding and porosity

`welding/arc.py` `air_fraction`. Air reaches the pool when the nozzle is too far from the work (beyond 1.5 nozzle diameters of laminar core, derived from the gas exit speed), when the torch leans beyond 22.5° travel angle (the trade's limit, read), in a cross draft from 1.4 m/s: 1 % air at 2.4 m/s and 1.5 % at 2.9 m/s (Kobelco's blowhole-against-wind chart, read at the cell's 19 L/min between its 15 and 20 L/min curves; the radiographic grades' mapping to air fractions is an assumption; `params.AIR_DRAFT`), or with the gas off (the arc then burns in air: air fraction 1). Porosity frozen into a solidifying slice grows from zero below 1 % air to 5 % at 2 % air (TWI: 1 % air entrainment gives distributed porosity, 1.5 % gross surface porosity, read; the curve between is phenomenological; capped at 15 %). Only the air under a burning arc counts: the nitrogen that makes the pores enters through the arc (dissociated in the plasma, carried by the droplets); a pool left open to air after the arc goes out takes up almost none (molecular N_2 at Sieverts' equilibrium, ~0.04 wt % in air, through a liquid-side mass-transfer coefficient ~10^{-4} m/s: under 0.001 wt % into a 3 mm pool in 0.5 s; the solubility and the coefficient are recalled textbook magnitudes, not re-checked against a source here, and the arithmetic is derived). A slice melted again (a fillet over a tack) starts its gas history again: its old pores melt out.

What losing the gas changes besides pores. The air under the arc dilutes the argon: below 82 % argon there is no spray transfer (§5), the arc runs globular and its electrode efficiency drops from 98 % to 89 %: more spatter (the spatter ledger, drawn in flight and stuck to the plate) and less weld metal. The bead's look follows the air under the arc (dark oxide, soot beside it; `render.py`) and its frozen porosity is drawn as open pores and in the macro sections. Not modelled: the arc's instability in air (wander, short-circuiting bursts) and the oxide's effect on wetting; no source with numbers for welding without gas was found.

The robot programs hold the torch over the crater with the gas on for a post-flow after the arc (0.5 s, the Miller Auto-Deltaweld default as Hirebotics integrates it) and then close the valve: it is what a real program does, and the oxidation it prevents is not modelled. The valve is closed at rest in every start state (a contract test checks all three tasks).

8. The bond between bracket and plate

`welding/bonds.py`. Per slice, the parts are bonded where solid weld metal (or melted and refrozen parent metal) fills the root and has fused into **both** parts; the slice's load path is the effective throat (the shortest distance through solid joint material from the root to the face). Each bonded section i carries a throat area a_i Δx_i k_y(T_i) at the fillet's ultimate shear strength τ = 0.852 F_EXX (AWS D1.1:2000 Commentary C2.14.4, test-based, Lesik 1990 for E70; F_EXX = 482 MPa, AWS A5.18). The codes' 0.6 F_EXX (AISC 360 J2.4) and 0.30 F_EXX (D1.1 allowable) are design values with margins, not the load at which a weld breaks. k_y(T) is the EN 1993-1-2 strength reduction at the section's hottest joint metal; a section liquid at its root carries nothing.

When it breaks: per load direction. A rule on the norms of the whole force and moment would break the bond at the same load whether the bracket is pushed into the weld or pulled off it. The bond's load is read in the weld frame at its anchor on the root line (measured: the reported force is the load on the bracket, the moment is about the anchor) and set against the weld group:

Load on the bracket	What carries it	Capacity
F_x, along the weld	longitudinal shear in the throats	C = τ Σ a_i
F_y, across the weld	transverse shear	f C, f = 1 (`WELD_TRANSVERSE_FACTOR`; below)
F_z > 0, pulling it off	transverse tension	f C; F_z < 0 bears on the plate
M_x > 0, top away from the weld side	peel: the bracket pivots on its far bottom edge, the welds pull on the lever a = t_b + throat/√2	f C a; M_x < 0 turns the bracket into the welds, which bear it in compression (not a weld failure)
M_y, lifting one end	the bracket pivots on the far end's bottom corner, the sections pull in proportion to their distance d_i from it	f τ Σ a_i d_i
M_z, twisting in plan	transverse shear about the group's centroid	f τ Σ a_i r_i, r_i the section's mean distance from the centroid over its own length (a single 0.5 mm section carries a Δx/4)

(plastic weld-group capacities: every section at its strength), combined linearly; the bond breaks when the sum of the six ratios exceeds 1. The rule is a hand calculation of the weld group (D1.1 2.25.2 asks designers to avoid tension at the root of a single fillet: the peel lever is a modelling choice, labelled derived, not a code formula; none was found); E7 compares the simulated break with its closed form for the peel case and shows the reverse direction holding.

No directional increase. The design codes' 1 + 0.5 sin^1.5 θ factor (1.5 across the weld; AISC 360 J2.4, AWS D1.1) was derived from fillets welded on both sides of the loaded plate. Forty single-sided fillets under transverse load broke at 0.765 F_EXX on the throat when loaded nominally concentrically, at 0.372-0.465 F_EXX when the load's offset bent the root open, and at 0.96-1.05 F_EXX when it bent the root closed, against 1.50 F_EXX double-sided; the authors find the factor unsafe for every single-sided fillet (Thomas & Tousignant 2022, J. Struct. Eng. 148(9) 04022118, Table 6, read: mean test-to-predicted ratios 0.761, 0.463 / 0.370, 0.955 / 1.043 and 1.493 against CSA S16 with the factor, a nominal 1.005 F_EXX). CSA S16:19 no longer allows it for single-sided fillets to elements in tension. Without the factor the rule's transverse strength, 0.852 F_EXX, is 11 % above the single-sided tests loaded nearly concentrically (their scatter is 12 %), 1.8-2.3 times the tests whose root is opened and 11-19 % under those whose root is closed: the real strength depends on which way the load bends the root, which one factor cannot represent. Peel opens the root, so in peel the bond remains on the strong side.

Shrinkage distortion. As bonded weld metal cools below the solidus, the bond's rest pose turns about the root line, the bracket's top toward the weld side, by 1.15° for a 22.7 mm² × 150 mm single-sided fillet (`SHRINK_ANGLE_REF`, derived), scaled by the cooled bonded throat² × length (the angular distortion of a fillet grows with its weld metal's cross-section; each section counts by the fraction of its cooling from the solidus to 20 °C that is done). The 1.15° comes from measured joints: free, tacked A36 T-joints with one GMAW fillet (La Porta Neto 2017, UFRGS, Table 5.1, read) closed their web-to-flange angle by 0.83-1.05° at 910 J/mm and 1.33-1.44° at 1385 J/mm on 9.5 mm flanges; this cell's procedure (1125-1179 J/mm, 21.7-22.7 mm² of deposit) lies between them, at 1.14-1.19° (the flange's own angle 1.19-1.24°). Their joints were unrestrained. Here the toggle clamps press the plate's back wing and the weld-side wing is free to lift, so they do not restrain this angle (about 50 N m about the root, against the ~700 N m that would hold that wing flat; derived). Their welds were short-circuit beads with 1.2 mm wire, welded flat; the specimens scatter by 0.1-0.3°. Satoh & Terasaki's bead-on-plate relation (~0.6°) and the AWS Welding Handbook's fillet chart extrapolated (~0.9°) lie under these measurements. The plate is rigid in the model, so the bracket turns; in the cell the weld-side wing would lift instead, with the same angle between bracket and plate at the weld. The MuJoCo weld equality pulls the rigid bracket to that pose; nothing sets the angle on the body. A bond made now rests at its present pose, so the shrinkage so far is in that pose already. E25 is a **tracking check**: the bracket body turned 0.117° toward the weld side against the loose bracket's fit-up pose, where the bond's target was 0.116°, and held within 0.0017° over 2 s hands-off [E25]. Its check that the target equals the law recomputes the same formula from the same sections, so it cannot fail for a wrong law: E25 shows that the bracket follows the bond, not that the angle is right; the angle rests on the measurements above. No law for tacks was found: the volume scaling

gives the two tacks about 0.1°, a scaling by weld size alone about 0.5°. Transverse shrinkage (about 0.8 mm per fillet, `SHRINK_TRANSVERSE`) is not built. The squareness check of the tack task sees the tacks' pull as well as how the bracket was held.

Mechanically the bond is one MuJoCo weld equality between the two bodies, switched on at first fusion at the parts' current pose. The sections bonded at a break are cracked: they carry nothing until they melt again; new fusion elsewhere bonds the parts again, but only while the bracket is within 1 mm and 2° of its fit-up pose (the thermal grid's weld metal is where the bracket was fitted). The equality's stiffness grows with the bond's capacity (both scale with the fused throat area and the steel's strength as it cools): 1.52 N m/rad per newton of capacity, from 19 600 N m/rad measured on a 12 918 N tack bond inside the 2×10^{3}-1.5×10^{5} N m/rad derived for a real tacked joint; the impedance for it comes from the soft-constraint law measured in this cell, $k = C_0/(d(1-d))$, $C_0 = 9.8$ N m/rad (fits 981 / 4905 / 19 604 N m/rad at $d = 0.99$ / 0.998 / 0.9995 within 1 %).

A loose bracket away from its fit-up pose is one object. The thermal grid lives in the base plate's frame and holds the bracket at its fit-up pose. While the bracket is not bonded and is more than that tolerance (1 mm, 2°) from the pose, its cells are out of the grid -- the arc, the wire and touch sensing meet the rigid bracket's collision box, the cameras draw the rigid body whole -- and they come back once it is within 80 % of the tolerance; their heat and volume are booked in the energy and metal ledgers meanwhile (`process.detach_part`), so the arc cannot weld to bracket metal that is not there. An arc on the moved bracket burns as a stray arc (§11.3): the source runs it and books its metal and heat, which the bracket's cells, being out of the grid, do not receive.

9. Robots and controls

- **UR10e** (torch arm): URDF masses, inertias and limits from Universal Robots' ROS 2 description (joint torque limits 330/330/150/54/54/54 N m from that URDF, since UR's datasheet publishes none; speeds 120/120/180/180/180/180 °/s, as in the URDF and UR's datasheet); position servos with authored gains (UR publishes none) and gravity compensation; a rate-limited motion generator at the joint speed limits and 4 rad/s² (prior; URScript's movej default is 1.4); the protective stop of its default safety configuration: the arm stops when the external force at its tool (the wrist links and the gun) or at its elbow (the upper arm and forearm) exceeds 150 N, where UR's force limit acts (UR e-Series manual, read), and then backs off along the last 50 ms of its path ("back off to a position where the force limit is not exceeded", UR manual SF7; the distance is authored).
- **FR3** (handling arm): the shared catalog's model with the filleting/label-application gains; libfranka's default collision thresholds (10 N m, 20 N m while accelerating) stop the arm.
- **Commands** (policy key): named joint targets, Cartesian pose/delta through the environment's IK (for the torch arm the Cartesian frame is the torch work point — on the wire axis 19 mm beyond the contact tip, z along the wire), the Franka Hand, and `tool_io`: the robot controller's outputs to the welding source (arc on, gas on, touch sensing, wire feed 1.3–25 m/min, voltage 10–44 V, wire inching forward/backward/stop at 2.5 m/min with the output off; no transfer-mode output, §5), validated before any side effect. **Feedback**: joint positions and velocities and the source's signals (arc on, current, voltage, actual wire feed, touch, fault, gas).
- **Deployments**: `ur10e-fr3` (default); `ur10e` (torch arm only; refused for the tack task, which needs the hand); `dual-fr3`: the right FR3 carries a **light torch** (`welding/torch.py RM2`: a 1.60 kg robot torch of the TBi RM2 class on a KS-2 class collision sensor, TCP 290 mm from the flange; the hose package hangs from a balancer and is not carried) inside the FR3's 3 kg payload, which the 2.76 kg cobot gun plus hose would exceed. E18 measures its static joint torques along the seam: at most 36 % of any joint's limit (the UR10e with its gun: 25 %). The FR3 torch arm's profile (`welding_torch`) has the same joint gains as the handling arm's, no hand, its own start pose, and the torch on link 8. `single-fr3`: that FR3 torch arm alone (refused for the tack task, like `ur10e`). A rig may also be given as its id, setup object or JSON path (`welding/robot_setup.py`). The robot and task are checked together at launch (§17).
- `dual-panda`: two Pandas at the table's back corners like the dual-FR3; the right Panda without its Franka Hand (the pinned Panda package has no bare-arm URDF: the hand and its fingers are stripped from `panda_arm_hand.urdf` at load) carries the same 1.6 kg light torch on link 8; the left Panda has the Franka Hand; libfranka's default collision thresholds. E18: at most 32 % (joint 2). Its tack program holds the bracket's left end and tacks the right end first with 15 N per finger: the default order swings the hand across the torch arm and both arms stop on a ~2100 N arm-to-arm contact (measured), and at 30 N per finger the Panda reads 12-16 N m at joint 3 against the FR3's 4.3 N m for the same grip and trips its 10 N m reflex (the reading is a one-sided finger push while the hand closes: the Panda's right finger meets the bracket first, its left closes on without touching, and 25.9 N on one finger reads 12.8 N m at joint 3; the FR3's fingers meet the bracket together and read 0.14 N m. The two fingers are coupled by MuJoCo's default soft equality (20 ms), which on the Panda's 14 g fingers lets them part by millimetres; at 4 ms the peak falls to 8.6 N m without a reflex and the fingers stay together. The remaining difference is not determined).
- `g1`: the Unitree G1 humanoid standing 0.10 m behind the table on its position servos with Unitree's gravity feed-forward on the waist and arms (as its manipulation stack sends; no balance controller, and its low-level joint interface has no collision or speed reflex), legs sat back and the waist leaning 0.2 rad: its first stance fell (centre of mass 70 mm in front of the ankles), this one stands with at most 1.6° of tilt. The light torch is clamped to the back of the right Dex3 palm (0.15 kg aluminium saddle, authored geometry); the left Dex3 is commanded joint by joint (no `gripper` action). E18: waist pitch 62 % of its 35 N m (it holds the leaning torso and both arms), right shoulder pitch 53 %, wrist yaw 43 % of its 5 N m. The torch reaches the whole seam at the procedure's angles with the arm alone. Its arm program for the torch moves smoothly and holds 2.5 s still after each move: the robot sways on its leg servos, and a program that chases the sway puts the wire 3.2 mm from the root. **The tack task is not offered on the G1** (refused at launch, `variants.check_robot`). Reaching with the left hand to hold the bracket's top edge by a joint-space move sweeps the left elbow onto the table top and the robot falls: its centre of mass goes 63 mm outside the feet, and it ends at 20 degrees, propped on its wrist. Balanced press postures exist on paper (40 of 46 at x = 135 mm, the centre of mass up to 57 mm inside the feet), and a reach planned over the table keeps the robot standing. But the Dex3's thumb hangs 42 mm below the palm, so the palm does not reach the edge: the thumb lands on the plate, or, folded, takes the load itself (5-9 N). The posture that also lets the torch reach the tack leans the torso on the table (17-26 N), and the torch still ends 62 mm from the tack's start. A program that solves the hold-above posture (palm 50 mm over the edge at x = 135 mm, waist and arm) and the torch's pose 120 mm back from the first tack moves there, then finds no press pose for the left arm with the waist where it stands (palm face 4 mm into the edge, facing down within 25 degrees) and stops before any arc.
- **The G1's start state** is settled before handover until its torch, its hand and the coupon move less than 0.025 mm in a second: 13.3 simulated s from the import pose, done once per physics version and cached; later starts settle 1.3 s. It then stays at rest: E0 holds it 120 s, and its torch moves at most 0.0156 mm (tack start) and 0.0114 mm (fillet-2f start), with no steady drift (§15).

- The dual-Panda and G1 deployments weld the fillet as the UR10e and the dual-FR3 do: 8 of 11 checks at Lincoln's procedure (all four deployments fail leg size, overlap over the first tack's start, and the crater at the specified size), position-torch 8 of 8, and the dual-Panda's tack 7 of 9 (its tacks fail D1.1's tack checks on convexity and size); no protective stop in any of them (§14).

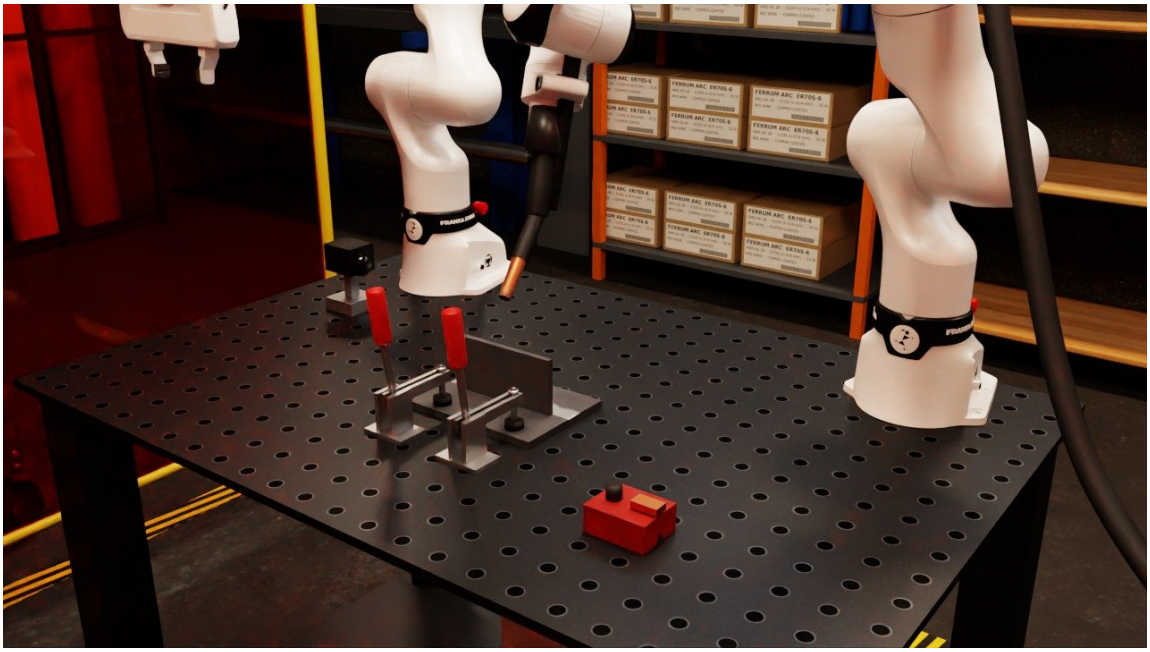


Figure 9.1 — The dual-FR3 deployment, tack start state: the right FR3 carries the 1.6 kg light torch, the left FR3 the Franka Hand.

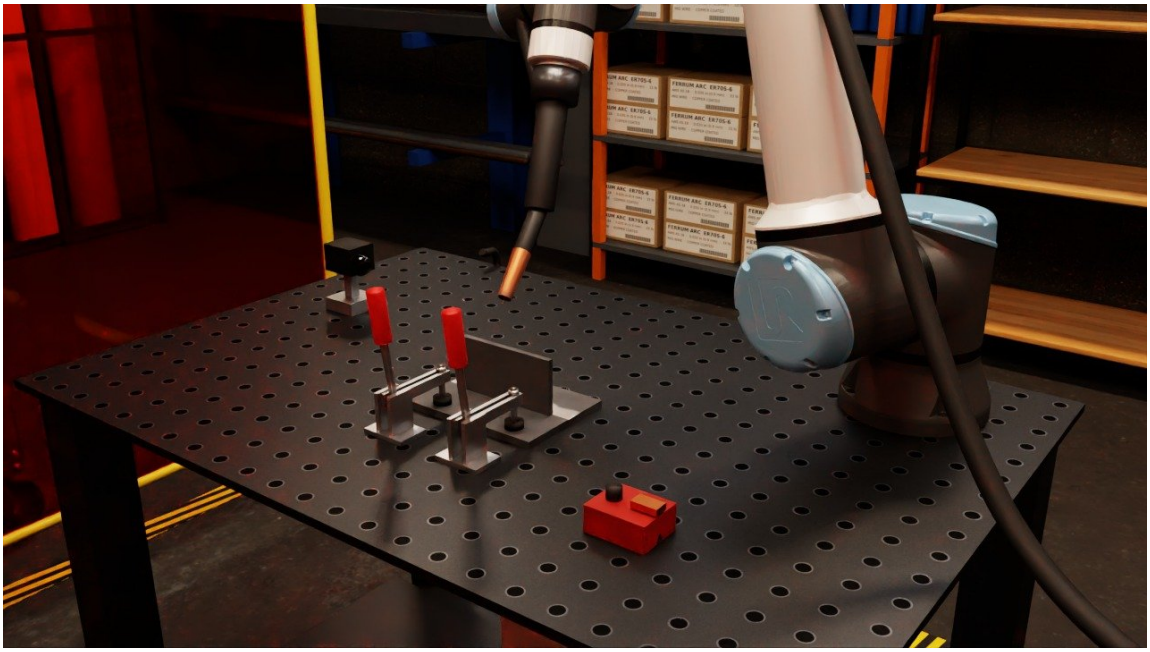


Figure 9.2 — The torch-only UR10e deployment (fillet-2f start state).

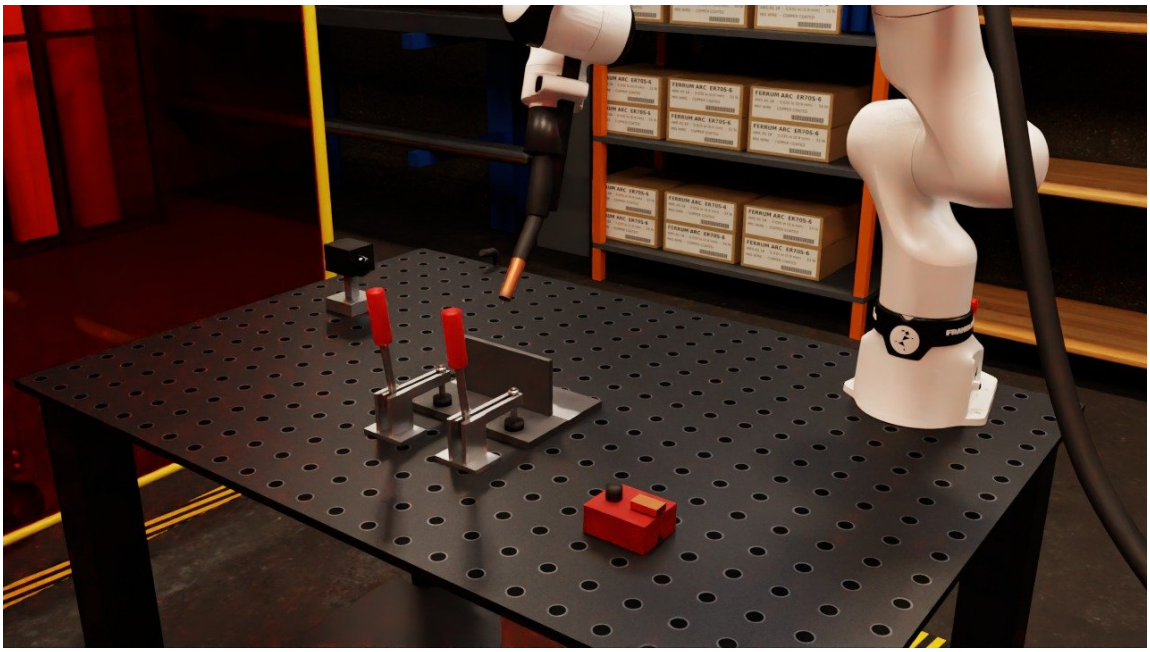


Figure 9.3 — The single-FR3 deployment: the light-torch FR3 alone (fillet-2f start state).

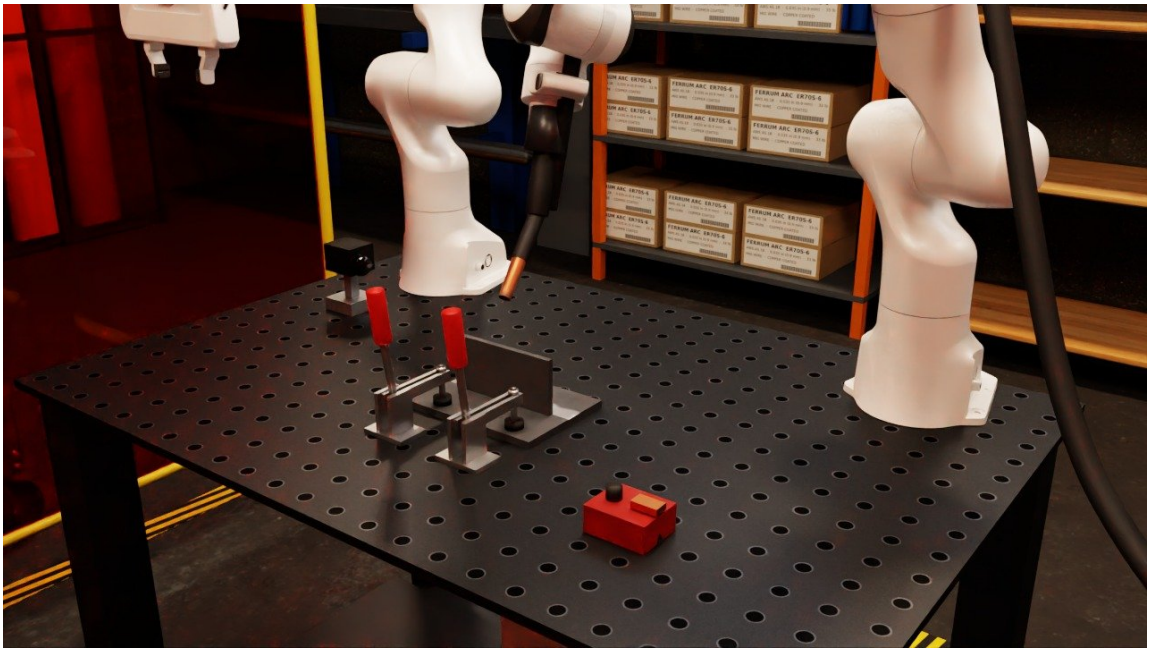


Figure 9.4 — The dual-Panda deployment (tack start state): the right Panda with the light torch on its flange and no hand, the left Panda with the Franka Hand.

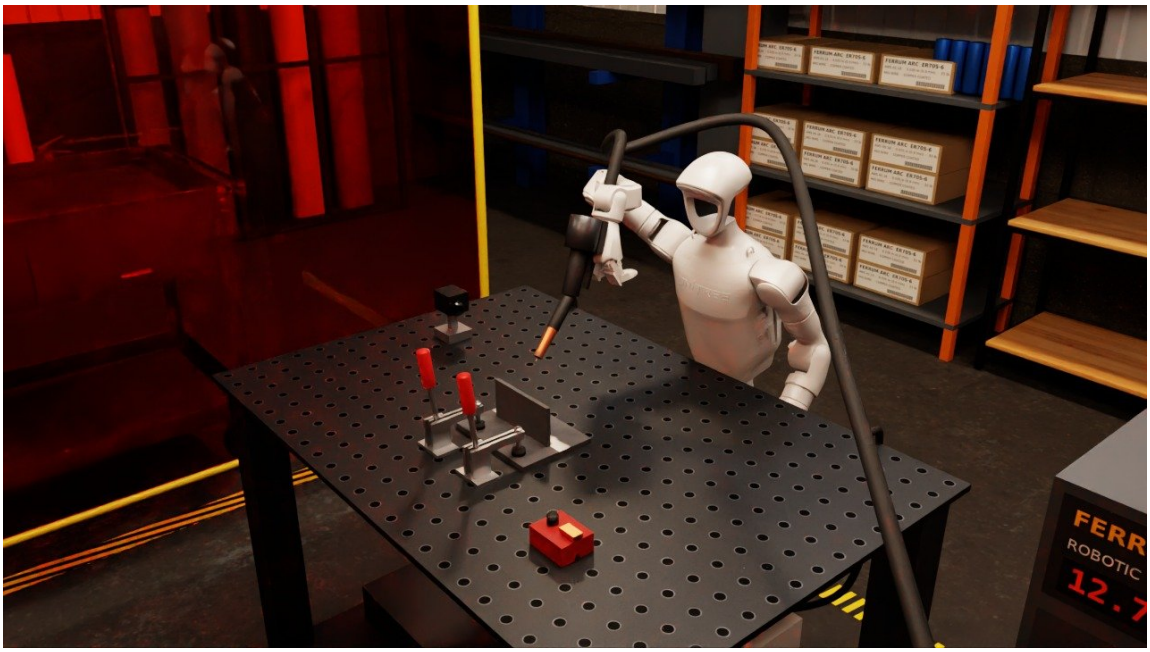


Figure 9.5 — The G1 deployment (fillet-2f start state): standing behind the table, leaning, the light torch clamped to the back of the right hand; the hose over its back.

10. Parameters

The full ledger, every constant with value, unit, provenance label, basis and the module that reads it, is `welding/params.py`: 78 model parameters, all read by the model, and 47 reference-only (the unbuilt presets B and C, superseded values and not-built items). Besides the labels above, `secondary` marks a value quoted in another source that was opened, the primary not.

Provenance notes. The wire-melting constants are this build's own fit of Lesnewich's form to the procedure charts (`LESNEWICH_A/B`, derived), because Lesnewich's published values could not be read. The arc's constants are Golob & Tovornik's set, used unfitted, and the stick-out's resistivity is anchored to measured ER70S-6 data. The liquid's enhanced conductivity factor is an estimate (the 420 W/m K is read from the Penn State conduction-only models; its use for this wire and current is not), and the arc's heat spread σ is open, with its sensitivity measured by E3. Read: the 0.035 in wire's spray transition at 180 A in 90/10 (Lincoln C4.200), the 82 % argon limit for spray, the electrode efficiencies by mode, the ultimate fillet shear 0.852 F_EXX (D1.1 C2.14.4). Estimate: the short-circuit arc length (one wire diameter, 0.889 mm).

11. Component validation

Each experiment compares a part of the model with an independent answer (an analytic solution, a published procedure, a geometric distance or the model's own predeclared rule) and states its check before running. All ran on this build.

11.1 Heat (E1: PASS)

Test	Model	Reference	Error
Energy: insulated block, 500 W for 5 s	stored 2500.0000038 J	2500.0000058 J in	-8.1×10^{-10} relative (-8.9×10^{-8} without the compensated enthalpy)
Stefan: one-phase Neumann solidification (Ste = 2.35)	front 4.933 / 6.976 / 9.852 / 13.940 mm at 1 / 2 / 4 / 8 s	4.931 / 6.969 / 9.855 / 13.937 mm	≤ 0.11 %
Rosenthal thick plate, 1 kW at 5 mm/s (σ = 1.5 mm, 0.5 mm cells)	849 / 521 / 352 / 219 °C at 5 / 8 / 12 / 20 mm behind the arc	852 / 531 / 358 / 221 °C	−0.3 to −2.0 %; 400 °C isotherm half-width 4.25 vs 4.39 mm (−3.2 %, one 0.25 mm cell is 5.7 %)

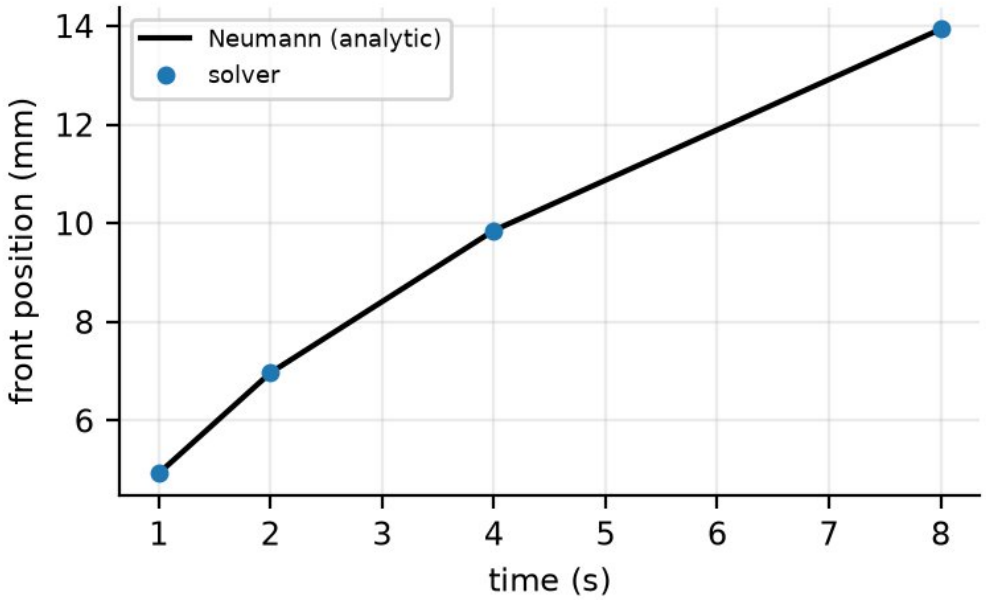


Figure 11.1 — E1: the solver's solidification front against the analytic Neumann solution.

11.2 Arc (E2)

The source model alone, on the CPU, at the chart's stand-off. The transfer mode is the model's (§5), set against the chart's:

Schedule	WFS, V, CTWD	Chart	Model	Role
spray, 0.035 in, 90/10	12.7 m/min, 29 V, 19 mm	230 A, spray	241.3 A (+4.9 %), spray	in the melting fit; no arc constant fitted
short circuit, C25	6.4 m/min, 20 V, 12 mm	175 A, short circuit	163.4 A (−6.6 %), short circuit	in the melting fit
short circuit, sheet	3.8 m/min, 17 V, 12.7 mm	120 A, short circuit	112.6 A (−6.2 %), short circuit	in the melting fit
short circuit, vertical	5.7 m/min, 18 V, 12 mm	160 A, short circuit	151.3 A (−5.4 %), short circuit	prediction: in no fit

What is independent. No arc constant is fitted (Golob & Tovornik's set, §5); the wire-melting law's two constants are fitted to the spray, 175 A and sheet rows; only the vertical row was in no fit, and it is the one prediction: −5.4 %. The modes are not fitted anywhere: all four come out as the chart says. The short-circuit wire feed and voltage (6.4 m/min, 20 V) at the spray stand-off of 19 mm is short circuit at 137 A; the spray schedule with the gas off (air under the arc) is globular at the same 241.3 A; with 1.8 % air it stays spray (88 % argon). Over a map of 200 operating points at 90 % argon (wire feed 3.8-20 m/min, 16-32 V, 12 and 19 mm) the model gives 86 short circuit, 22 globular and 92 spray (`mode_map`).

Self-regulation: at the spray schedule the current falls 6.9 A per millimetre of contact-tip-to-work distance (13 to 26 mm: 292.9 → 202.7 A), steeper than the 3.9-4.8 A/mm measured for 1.32 mm wire in spray (Kim, Zhang & DebRoy 2003 Table I) by 45-78 %. In the model the slope is the arc's field over the loop's resistance plus the melting law's coupling: at the procedure 0.030 Ω arc, 0.006 Ω cable, 0.018 Ω stick-out and 0.043 Ω of coupling give −6.94 A/mm. It is an outcome of the constants, not tuned; every published arc constant set gives 6.8-7.6 A/mm. A static source slope of 4.3-7.6 V per 100 A would give the measured slope, but none was found for this source (`CV_SLOPE` = 0, estimate), so the difference stays disclosed. A 3 mm step in stand-off (19 → 22

mm) moves the current from 241.3 to 222.6 A with a 90 % settling time of 0.125 s, the order of the stick-out's burn-off time L_s/w_f (12.5 mm at 212 mm/s: 0.06 s, derived).

The source's limits (E2 (e), the source alone stepped at 1 ms for 1 s after the strike at 19 mm; §5):

Settings	What happens	Peak current	Wire balance
12.7 m/min, 29 V (the procedure)	spray, no fault	267.8 A (just after the strike's burn-off; 241.3 A steady)	0
1.3 m/min, 44 V	**burnback** 0.26 s after the strike: the arc burns the stick-out to the contact tip; 11.3 mm³ deposited = the 6.1 mm³ fed plus the stick-out it melted, less spatter	400 A (at the ceiling)	0
12.7 m/min, 16 V	**stubbing** at the strike: the wire drives into the work, the drive pushes the gun back 48.2 N (the 17.7 mm stick-out's buckling load)	—	0
25 m/min, 44 V	runs **at the 400 A ceiling**, the source's voltage below its set point	400 A	0

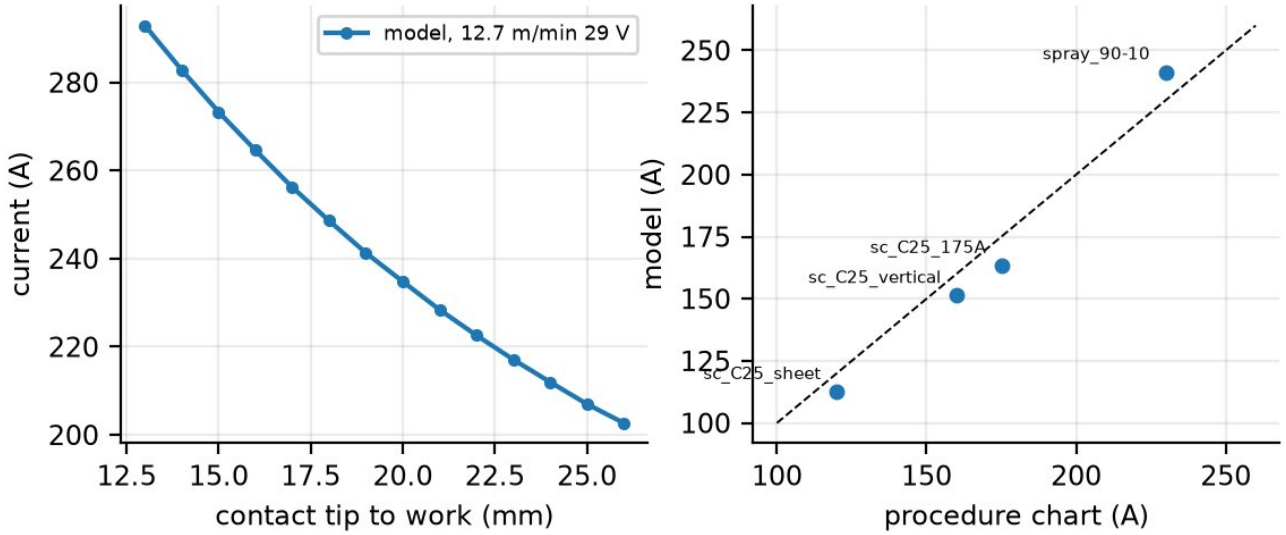


Figure 11.2 — E2: current against contact-tip-to-work distance at the spray schedule (left); model against the procedure charts (right; the two prediction rows were not fitted).

11.3 Welding circuit, wire and touch sensing (E6: PASS; SCRIPTED ROBOT)

The circuit, from the cell: the magnetic ground clamp stands on the steel fixture table, so the table, and the zinc-plated toggle clamps bolted to it, are at work potential, and so is everything resting on them with load (the base plate, and the bracket through it or through the bond). The power source's case is not in the welding circuit (a welding source's output is insulated from its case); the robots and the painted equipment are not conductors here. The UR10e brings the torch over the seam start with the IK; the wire is first inched from the 11.5 mm left by the tack fixture to 20 mm with `wire_inch` (the tapered nozzle's 19 mm end clears both parts in this corner from 12.7 mm contact tip to work: its end must be at least its 9.5 mm radius from the root, plus the 3.2 mm tip recess; measured).

Case	What happened	Check
(a) Arc on with the ground clamp parked on the source's case (no circuit)	the wire inched until it met the tack and stopped there (15.1 mm stick-out = the stand-off); no current flowed; the source reported `arc_start_failure` after its 2 s	no arc, fault raised, wire ends at the metal: PASS
(b) Touch sensing down the wire axis, clamp on the table	contact reported at 20.023 mm stick-out; the geometric distance from the tip to the root along the axis was 20.194 mm (−0.17 mm, within one cell and one 0.1 mm search step); no protective stop, no gun contact	contact within 0.35 mm: PASS
(c) Same, clamp on the source (no circuit)	no contact reported; the search continued until the nozzle met the plate and the bracket (62 N and 42 N, below the UR10e's 150 N stop)	no false contact: PASS
(d) The wire off the coupon	over the bare table the inched wire stopped on the top (53 mm stick-out) and touch sensing reported contact. With the output on, a **stray arc** struck there and burned 2 s at the source's operating point for a contact tip 53 mm from the table (globular, 145 → 136 A on a 49 mm stick-out); the coupon received no metal and no heat; 235 mm³ of wire metal and 6.3 kJ were booked as lost onto the table, and the wire balance closed to 10^{-12} mm³. Over a toggle clamp's steel arm: the same, 1 s (118 mm³, 3.2 kJ). Over the power source's case top: the wire stopped on it, touch sensing read nothing, and the output on gave no arc and `arc_start_failure`. Pointing into the room: inching stopped at the 100 mm bound and the output on gave no arc and `arc_start_failure`. No protective stop	stray arcs on grounded steel only, books closed: PASS

Case (c) is the physical consequence of an open circuit during a touch search: the wire does not sense, so a search that does not bound its own travel ends with the nozzle on the part. Case (d) is the trade's stray arc: a wire that touches grounded steel anywhere strikes there. The stray arc's metal and heat are booked, not modelled on the body it burns on (§18).

11.4 Torch arm loads (E18: PASS)

Static joint torques with the arm holding the torch at the procedure's angles along the whole seam (IK solved at each point, gravity compensation off, so the numbers are what the motors carry): the UR10e with its 2.76 kg gun uses at most 25 % of any joint's rated torque; the FR3 with the 1.6 kg light torch at most 36 % (joint 2, 31 N m of 87 N m, at the seam start), its wrist joints (12 N m) at most 14 %. The light torch's own share of the FR3's shoulder torque is 4-6 N m. The 2.76 kg gun on an FR3 would leave 0.24 kg of its 3 kg payload for the hose package, which is why the dual-FR3 deployment carries the light torch and hangs the hose on a balancer. The dual-Panda uses at most 32 % (joint 2, 28.1 of 87 N m; wrist joints at most 6.8 %), and the G1, at most 62 % (waist pitch, 21.5 of 35 N m; arm: shoulder pitch 53 %, of which the torch is 6.1 N m; wrist yaw 43 % of 5 N m, the torch 1.8 N m); a stance 0.55 m back with a 0.3 rad lean would load the G1's waist pitch to 73 %, over the 70 % rule, so it stands as in §9. The worst joints are: dual-fr3 35.7 %, ur10e-fr3 25.4 %, dual-panda 32.3 %, g1 61.6 %.

11.5 The hand holds the part (E19; SCRIPTED ROBOT)

On the tack task's start the FR3 grips the bracket's top edge 120 mm along it with the Franka Hand in force mode (30 N per finger), lifts 20 mm and holds it 1 s. The fingers stop on the bracket's faces (half-openings 3.78 and 3.84 mm for its 8 mm thickness), the contact forces sum to 60.0 N, and the bracket rises 15.88 mm with the hand's 15.5 mm (the arm sags 4.5 mm under the part's weight with the catalogue's joint gains; slip −0.38 mm), no protective stop. The probe also records a check of MuJoCo's bounding trees: 26 collision geoms lie outside their body's MuJoCo bounding tree (the FR3 links by up to 171 mm, the UR10e's upper arm and forearm by up to 596 mm, the torch's nozzle by 110 mm and neck by 57 mm, the coupon's boxes and the clamp pads by 1 mm), harmless only because the midphase that uses those trees is off.

11.6 The tack bond (E7; ASSISTED phantom force)

The fillet-2f start state's two tacks are peeled by a phantom force on the bracket's top edge (middle of its length, 75 mm above the root, straight away from the weld face), ramped at 2 kN/s, with the equal and opposite force on the base plate (held as in a vise); both forces end at the break. The same start state is then pushed the other way, into the weld, to twice the peel break.

Three predictions. (1) The bond rule's own closed form, $P^* = f\,C / (1 + h/a)$, with $f = 1$ (§8), C the group's capacity and $a = t_b + \text{throat}/\sqrt{2}$. This checks the implementation, not the mechanics. (2) The inspector's hand calculation: the tacks measured on the evaluator's sections (fused at the root; size; effective throat 0.707 × size; two tacks, 94.6 mm² of throat area), the same pivot lever (10.23 mm) and the rule's strength, 0.852 F_EXX. It shares the rule's strength, directional factor, interaction and lever and differs only in the throat measure, so it tests the throat measure, not the rule. (3) Sharing only the pivot lever: the welds' transverse strength taken from tests of single-sided fillets (Thomas & Tousignant 2022, read). Those are 4186 N for nominally concentric specimens (0.76 F_EXX on the throat), 2550 / 2037 N with the root opened by a 15 / 30 mm offset, 5253 / 5737 N with the root closed by the same offsets (scaled from the concentric value by the paper's ratios, 0.955 / 1.043 against 0.761), and 8221 N from double-sided specimens (1.50 F_EXX).

Case	Capacity C	Rule's closed form	Hand calculation	Single-sided tests	Break
cold tacks (cooled 8 s)	44 109 N	5434 N	4666 N (ultimate); 3286 N at the design strength 0.6 F_EXX	2037-4186 N (double-sided 8221 N)	5438 N (closed form +0.08 %; hand calculation ×1.17; concentric single-sided tests ×1.30)
cold tacks pushed toward the weld side to twice the peel break (10 876 N)	44 109 N	— (bearing)	—	—	none; the bracket tilts 0.7° (the bond's stiffness law)
hot tacks (0.3 s after the arc, joint metal up to 1488 °C)	18 787 N	5434 N	—	—	5438 N

The rule has no directional factor (§8) because with the codes' 1.5 the same tacks break at 7632 N, where a double-sided fillet of this throat would break (0.99×), 1.9× above the concentric single-sided tests and 3-4× above the root-opening ones. The break lies 30 % above the concentric single-sided tests (their scatter is 12 %) and at about twice the tests whose root opens. Peel opens the root, so the bond remains on the strong side. The simulated break is 17 % above the hand calculation because the model's throat is the shortest cut through solid joint material in each section, including the root's penetration and the face's convexity, while the inspector's 0.707 × size is not. The hot tacks start at 43 % of the cold capacity (k_y at their temperature); the ramp lets them cool, and they break at the cold load. The bracket is rigid in the model. At the break its moment at the tacks, about 0.38 kN m, stays under the first yield of a 150 × 8 mm A36 bracket (0.40-0.56 kN m for F_y

250-350 MPa), so a real bracket would still be elastic here. **The film shows the break.** After the measured window, an ASSISTED display pull on the bracket alone (40 mm/s, at most 2 N, no reaction on the plate, which its clamps hold) tips the freed bracket over its far edge onto the clamp arms. The tack metal stays on the plate. The numbers above are recorded before it.

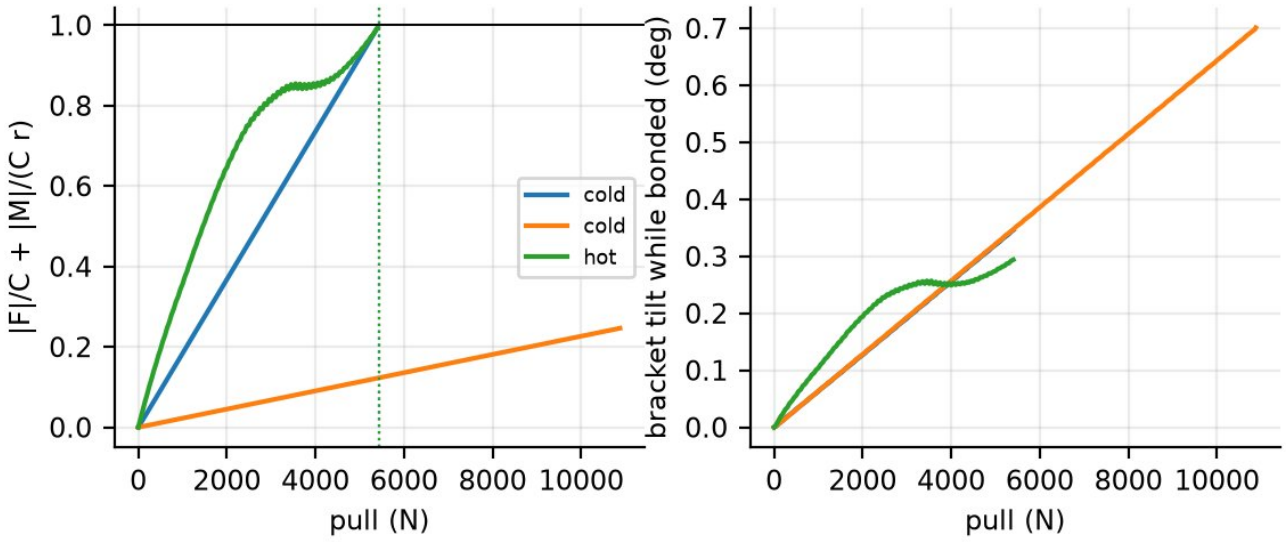


Figure 11.3 — E7: the interaction ratio and the bracket's tilt against the phantom pull while bonded; dotted: the predicted break.

11.7 The fillet under the nominal procedure (E4; ASSISTED torch path)

The whole process chain without a robot: a kinematic torch follows the joint at the Lincoln ¼ in 2F spray procedure (0.035 in ER70S-6, 90/10, 12.7 m/min, 29 V, 19 mm contact tip to work, 5.93 mm/s, 45° work angle, 7.5° push) over 130 mm from x = 10 mm, fills the crater 0.5 s and cools 5 s; the source is commanded as a robot controller would command it. Nothing is tuned to the result. The prediction is the procedure's: a ¼ in (6.35 mm) fillet, continuous, fused into both members, without undercut or porosity.

Quantity	Model	Reference
wire fed / spatter / deposited	2951 / 60 / 2892 mm³	spatter 2 % (spray electrode efficiency 98 %, Lincoln C4.200)
metal added to the cells	2891.6 mm³	the deposit exactly (mass-conserving shrinkage, §6); energy ledger residual 3.6 × 10⁻⁸ of the 103.1 kJ input; wire balance 0
legs bracket / plate (median, x = 25-125 mm)	6.25 / 6.25 mm	6.35 mm (¼ in): −1.6 %
fillet size, throat (median)	6.20 mm (6.02-6.20), 4.60 mm	6.35 mm, 4.49 mm
face convexity (max)	2.13 mm	AWS D1.1 Fig 5.4: ≤ 3.2 mm for a face wider than 7.9 mm
continuous weld	132.5 mm (x = 10.0-142.5 mm)	the 130 mm path from x = 10 mm
fusion into both members	every section of the continuous run (penetration at least 0.5 / 0.5 mm)	—
overlap (cold lap)	up to 2.5 mm, in sections between the 2 mm samples the result file keeps (none in any kept section; its place is not recorded)	D1.1: none
undercut, porosity	at most 0.75 mm (none over 1 mm), 0	D1.1: undercut ≤ 1 mm

The deposit makes a slightly convex 6.2 mm fillet where the procedure promises a flat 6.35 mm one: with the face's convexity the same metal covers less leg. Real welds at this procedure are also usually convex, so the 2.4 % short size is a disclosed model difference, not a calibrated one (§18). Simulated 27.6 s took 67 wall seconds per simulated second without rendering, on a GPU shared with renderers.

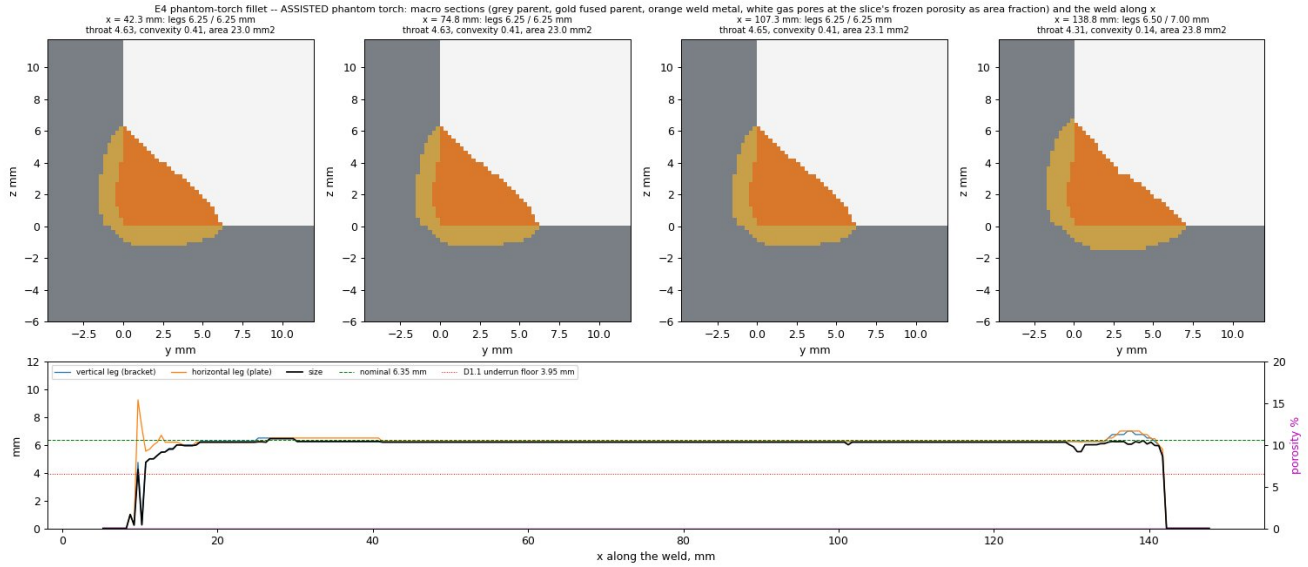


Figure 11.4 — E4: macro sections at four positions (grey parent metal, gold fused parent metal, orange weld metal) and the legs and size along x (green: the ¼ in nominal; dotted: the D1.1 underrun floor).

11.8 A bead on plate against a published section (E3; ASSISTED phantom torch)

E3 compares penetration and bead width with a real weld. The reference is a spray bead on 9.5 mm carbon steel with its macro section (Souza, Resende & Scotti, COBEF 2007): 1.2 mm ER70S-6, Ar + 2 % O_2, 250 A, 28.5 V, 5.8 mm/s, 20 mm contact tip to work; width 9.4 mm, reinforcement 3.0 mm, penetration 3.3 mm, fused parent area 16.8 mm², deposit 23.4 mm² per mm (derived). No published section at this environment's 0.9 mm procedure was found, so E3 holds the voltage, travel speed, stand-off and either the current or the deposit per mm, and changes the wire to the model's 0.9 mm: a check of the heat and pool at a matched operating point, not an identity (a 1.2 mm wire at 250 A has a lower current density). Measured on the steady 20 mm in the middle of a 40 mm bead; nothing is tuned to the result. Cases: the wire feed that gives the reference's 250 A (more metal than the reference: the 0.9 mm wire melts faster per ampere), the wire feed that deposits the reference's 23.4 mm² per mm, and, deposit matched, the arc's heat spread σ at 3 and 5 mm (4 mm in use; open), the liquid's conductivity at half and twice the Penn State 420 W/m K (estimate), and the gravity-consistent pool cap (§6; in use).

Case	σ mm	liquid k W/m K	cap	current A	width mm	reinforcement mm	penetration mm	fused mm²	deposit mm²
Souza et al. 2007 (1.2 mm wire, 250 A)	—	—	—	250	9.4	3.0	3.3	16.8	23.4
current_matched	4	420	solved	245	10.00 (+6 %)	2.75 (-8 %)	3.00 (-9 %)	17.3 (+3 %)	23.7
deposit_matched	4	420	solved	232	10.50 (+12 %)	1.75 (-42 %)	2.50 (-24 %)	15.6 (-7 %)	13.0
sigma3	3	420	solved	234	11.00 (+17 %)	2.00 (-33 %)	2.75 (-17 %)	19.8 (+18 %)	16.7
sigma5	5	420	solved	231	9.50 (+1 %)	2.38 (-21 %)	2.25 (-32 %)	11.9 (-29 %)	20.2
k_half	4	210	solved	232	11.00 (+17 %)	1.75 (-42 %)	2.50 (-24 %)	14.9 (-11 %)	13.0
k_double	4	840	solved	232	10.50 (+12 %)	1.50 (-50 %)	2.25 (-32 %)	15.4 (-8 %)	13.3
cap_solved	4	420	solved	232	10.50 (+12 %)	1.75 (-42 %)	2.50 (-24 %)	15.6 (-7 %)	13.0

(The median of the sections over the bead's middle 20 mm. `cap_solved` repeats `deposit_matched`.) **E3 fails: the bead on plate humps.** The deposit-matched bead alternates every 3-5 mm between flat slabs (8-12 mm² of deposit, 1.2 mm high) and domes (30-46 mm², 4-5.5 mm high: the figure's middle section is one), so the median section holds 13.0 mm² against the reference's 23.4 and reads 42 % low on reinforcement, while the metal is exact (939.5 against 938.5 mm³ over every section). The σ and conductivity variants hump the same way (their medians 13.0-16.7 mm²); the current-matched bead's median, 23.7 mm², lies between its slabs and domes and does not make it a match. The cause of the humps is not established (§18). The fillets on the tee do not hump (§14, §18). Ledger residuals 3.0-3.8 × 10^{-8}; 13-18 wall minutes per case on the shared GPU. No published section at the environment's own 0.9 mm wire in 90/10 gas was found, so the reference still differs in wire, gas and source type.

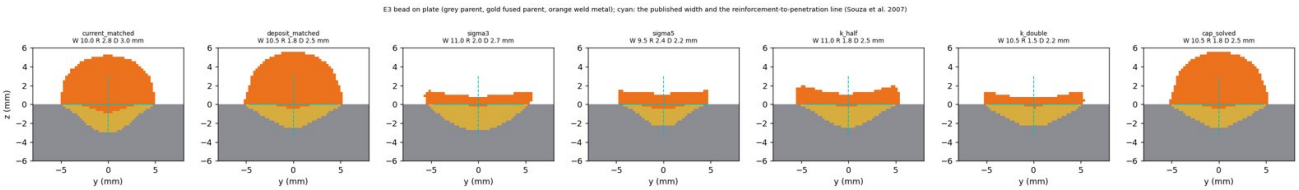


Figure 11.5 — E3: the middle section (x = 30 mm) of each case (grey parent metal, gold fused parent metal, orange weld metal); cyan: the published width and the reinforcement-to-penetration line. The beads hump: these are single sections of beads that alternate between slabs and domes along their length.

11.9 A destructive break of a robot-laid fillet (E14; SCRIPTED ROBOT weld; ASSISTED load)

The UR10e lays the fillet with the scripted program at 5.2 mm/s on the tacked start state, the cell cools and the unmodified evaluator judges it; then the joint is broken as AWS D1.1's fillet weld break test loads it, "in such a way that the root of the weld is in tension" (D1.1:2000 4.30.4): a phantom force on the bracket's top edge pulls it away from the weld side, reacted on the base plate as in a vise, ramped at 10 kN/s. Recorded: the evaluator's verdict, the break force against the weld group's closed form, and the test's acceptance read on the cells (D1.1 passes a break that shows complete fusion to the root along the welded length).

Quantity	Model	Reference
the weld (evaluator)	10 of 11: size ≥ 6.35 mm on 98 % of the length, throat 4.94 mm, crater filled to 6.47 mm, continuous 3.0-150 mm, fused at every macro section; fails overlap (up to 0.75 mm over 5.5 mm, over the first tack's start, §18)	the work order's checks (§13)
fused at the root, along the whole welded length	yes: all 280 sections with weld metal fused at the root and into both members	D1.1 4.30.4: a break that shows complete fusion to the root passes
bond capacity in longitudinal shear	306.4 kN over 147.5 mm	—
break load, top edge (75 mm, peel)	**40.54 kN**	the rule's closed form at the capacity in force at the break 40.64 kN (−0.2 %; taken before the pull 36.74 kN, the weld still cooling); the inspector's hand calculation from the macro sections (672.4 mm² of effective throat, lever 11.25 mm) 36.01 kN at the ultimate strength (×1.13), 25.36 kN at the design strength; single-sided test strengths 15.7-32.3 kN (concentric 32.3 kN: ×1.25)
break load, a press 8.5 mm above the root	**177.6 kN**	the closed form 179.7 kN (−1.2 %); the hand calculation at that height 157.3 kN (×1.13); single-sided test strengths 68.7-141.1 kN (concentric: ×1.26; double-sided 277.1 kN)

Taken before the pull, while the weld is still cooling and its capacity still rising (here from 277 to 306 kN during the pull), the closed form reads low; at the break it agrees within 0.2 %. The break lies 13 % above the inspector's hand calculation, because the model's throat (the shortest cut through solid joint material) counts penetration and convexity that 0.707 × size does not, and 25-26 % above single-sided fillets tested concentrically (Thomas & Tousignant 2022), as in E7. **The bracket is rigid here.** A real 150 × 8 mm A36 bracket loaded at its 75 mm top edge first yields at its weld toe at 5.7-8.0 kN (F_y 250-350 MPa), a fifth to a seventh of this break: it would bend over before the weld broke, as the D1.1 break test's stem "bends flat upon itself". So the 75 mm case checks the rule's implementation on a robot-laid weld and the D1.1 acceptance (fusion to the root); it does not predict a shop test. The press near the root is the shop test's loading. Even there the moment at the toe (about 0.6 kN m) is at a real bracket's first yield (0.40-0.56 kN m), so a real bracket would be at the edge of yielding when the weld broke. A weld that fails the evaluator's profile checks (here overlap) can still pass the destructive test, as on the shop floor: the break test judges fusion at the root, the visual inspection judges the profile.

11.10 A loose bracket away from its fit-up place (E24)

E24 pushes the tack task's loose bracket over with a phantom force and strikes an arc where it stood, and lifts it with the hand and sets it back:

Case	What happens	Measured
(a) knocked over (ASSISTED 2 N push on the top edge for 0.4 s), then the UR10e strikes at the seam's middle (SCRIPTED ROBOT)	the bracket tips 42.7° and comes to rest 27.6 mm from its place on the clamp arms; its 883 200 cells leave the grid	the arc's roots on weld metal only (body 255, never the bracket); metal added 133.91 mm³ against 133.90 mm³ deposited; energy ledger residual 3.6×10^{-8}
(b) the FR3's hand lifts it 20 mm (E19's 30 N grip) and sets it back, correcting by the bracket's measured error as an operator does by eye	out of the grid while up (16 mm off), back in when set down 0.20 mm off (0.21 mm after release)	ledger residual -7×10^{-9}; no protective stop

Case (b) passes only if the bracket returns to the grid, not merely if the grid and the bracket agree.

11.11 Weld metal as a collider (SCRIPTED ROBOT)

On the tacked start state the UR10e lays a 20 mm fillet (x = 60-80 mm) at the nominal procedure and the cell cools 4 s. The torch, its wire vertical and its neck turned away from the bracket, is then lowered at 2 mm/s with its nozzle's axis 14.5 mm out from the bracket face, so the nozzle's rim passes 4.95 mm out from it over the bead's slope; once over the bead (x = 70 mm) and once over the bare joint between the tacks (x = 45 mm).

Press	First contact	Nozzle end above the plate	Bead under the nozzle's rim (cells)
over the bead (x = 70 mm)	`gun_nozzle` on `weld_metal_035` (x = 69-71 mm)	2.45 mm	2.13-2.5 mm (median, maximum)
over the bare joint (x = 45 mm)	`gun_nozzle` on `base_plate_box`	0.00 mm	no weld metal

No protective stop in either press. The twelve boxes over the 20 mm bead (x = 61-85 mm) are each 2 mm long. Over the steady bead their empty space above the metal (the error of a box against the convex bead) is 0.9-1.8 mm (3.8-7.1 mm²); over the bead's cold start (the first segment) 3.7 mm (18.9 mm²), and 5.1-6.5 mm (25 mm²) in two segments of its last 6 mm, where it tapers into the crater. The nozzle, pressed at x = 70 mm, stops on the box over the bead under it 2.45 mm above the plate, where the bead's cells under its rim stand 2.13-2.5 mm.

12. Discretisation

The weld's shape depends on three numerical choices: the joint band's cell size (0.25 mm across, 0.5 mm along the weld), the pool update interval (5 ms) and the robot tick (1 ms; the thermal solver substeps at its explicit limit inside it). E12 reruns the same phantom-torch fillet (60 mm at the nominal procedure) with coarser cells (0.35 / 0.7 mm) and with the pool updated every 2 ms, and compares the sections with the reference. A quantity is converged if the coarse/fine difference is under one reference cell (0.25 mm) or 5 %.

Results (60 mm of weld from x = 10 mm; medians over x = 15-55 mm):

Quantity	Reference (0.25 / 0.5 mm, 5 ms)	Coarse cells (0.33-0.35 / 0.7 mm)	Pool every 2 ms	Converged?
legs bracket / plate (median)	6.25 / 6.25 mm	5.90 / 6.25 mm	6.25 / 6.50 mm	plate yes (+0.25 mm at 2 ms, one cell); bracket −0.35 mm coarse (−5.6 %, one coarse cell)
size, throat (median)	6.20, 4.63 mm	5.90, 4.64 mm	6.25, 4.63 mm	yes (≤ 0.3 mm, about one coarse cell)
section area (median)	23.4 mm²	23.1 mm²	23.4 mm²	yes (−1.3 %, 0 %)
continuous weld	132.5 mm (x = 10.0-142.5 mm)	the 130 mm path from x = 10 mm		
convexity (max over the run)	2.13 mm	1.90 mm	2.15 mm	yes; the steady face (median 0.47 / 0.75 / 0.39 mm) within about one coarse cell
penetration into each member (min over x = 15-55 mm)	1.00 / 1.00 mm	1.04 / 0.69 mm	1.00 / 1.00 mm	yes (one coarse cell)
overlap (max, at the weld's cold start)	2.50 mm	4.87 mm	1.25 mm	no
metal / deposit; energy residual	1.0001; 3.6e-08	1.0001; -1.5e-08	1.0000; 3.6e-08	—
wall time per simulated second	65	47	114	—

The steady fillet is converged to within one cell in every measure (the legs are counted on the pool's surface, but the coarse grid's surface is sampled on its own 0.33-0.35 mm cells), and the continuous start and the convexity maxima agree too. The cold start's overlap is not converged: 2.5, 4.9 and 1.25 mm, set by where the first droplets land and where the bulge touches down (§6, §18), which depends on the cell size and on how often the pool is solved. The grid builder cuts the fine band at the coupon's faces, so the joint's faces lie on cell edges at any cell size (faces inside cells smear the weld metal into sheets up to the band's edges); a contract test fingerprints the default grid.

13. Tasks and evaluation

Eleven tasks in the `steel-cell` preset (`welding/task.py`, `benchmark.json`): the three below and the eight of "The eight-task set" at the end of this section; legacy seeds 0-9 and the variation catalogue below move the coupon on the table within ±3 mm and ±2° (`variants.py`, `variations.py`). The evaluator reads physical state only: the cells (sections, bond), body poses, MuJoCo contacts and the source's recorded history; it never reads the commanded path. The weld's outer shape is read on the pool's free surface and checked against the cells (below).

Task	Criteria (all must pass)	Basis
`position-torch`	wire tip ≤ 1 mm from the root line, 0-10 mm from the seam start; work angle 40-50°; travel angle 0-15° push; contact tip to work 17-21 mm; the gun's hardware touched nothing during the episode (the wire may, for touch sensing); no arc; the TCP moved < 0.5 mm in the last second	angles and stand-off from the procedure (Lincoln C4.200, Miller, Bernard); tolerances authored
`tack`	exactly two tacks (fused stretches less than 10 mm apart are one tack), one within 40 mm of each end, each with 12.7-25.4 mm of fused length (½ in minimum, "never more than 1 inch", US Navy NAVEDTRA 14250); the bond active (a tack that cracked and was re-fused counts; the break stays in the bond's events); bracket square to the plate within 1°; within 1 mm of its fit-up position (the fit-up the thermal grid describes, the bond's own guard); the hand no longer touching it; the joint below 100 °C; every run of weld metal on the joint fused to both members along its length except one 0.5 mm section at each end, no overlap wider than one cell, convexity within D1.1 Fig 5.4, size ≤ 6.35 mm, porosity ≤ 1.5 %, undercut as the fillet's ("Tack welds shall be subject to the same quality requirements as the final welds", D1.1:2000 5.18.2; fusion Table 6.1(2))	read where cited (D1.1:2000); tolerances, one end section and the 6.35 mm size cap authored
`fillet-2f`	one continuous fused weld from ≤ 5 mm of one end to ≤ 5 mm of the other; size (the legs of the largest isosceles right triangle inscribed in the section, D1.1:2000 Annex B) ≥ 6.35 mm on ≥ 90 % of the span the weld must cover (5-145 mm) and ≥ 3.95 mm everywhere there (D1.1:2000 Table 6.1(6): a fillet may be under the specified size by 3/32 in on no more than 10 % of the length; the leg up the bracket's face steps in 0.25 mm, so it must reach 6.50 mm, below); median throat ≥ 0.707 × 6.35 mm; convexity ≤ 3.2 mm where the face is wider than 7.9 mm, else ≤ 1.6 mm (D1.1 Fig 5.4); undercut ≤ 2 mm and over 1 mm on ≤ 50 mm, anywhere on the joint (D1.1:2000 Table 6.1(7)(A): 2 in in any 12 in; the 150 mm joint lies within one 12 in); no overlap: weld metal lying on parent metal it did not fuse no wider than one cell, 0.25 mm (D1.1 Fig 5.4 accepts none; ISO 5817 only at its lowest level D); overlap over the span and every crater; within 5 mm of an end and outside every crater, where a weld starts, overlap up to two cells (0.5 mm) is accepted (the cold start, §18); convexity everywhere on the joint; macro sections at 6 mm, ¼, ½, ¾ and 144 mm (between the end allowance and the start state's tacks, so they show the fillet's own fusion) fused at the root and into both members by one cell, 0.25 mm; porosity ≤ 1.5 % in every section of weld metal on the joint (the model's gas fraction per 0.5 mm slice held to ISO 5817 C's figure, an authored, stricter use: D1.1 Table 6.1(8) limits visible piping porosity, which the model does not show); every crater filled to the specified size: ≥ 6.35 mm over 10 mm centred on each place a pool froze last (no section of its own pass within 5 mm froze later; a pass is the run of sections inside the joint whose adjacent freeze times differ by ≤ 2 s), the window run on from the stop to the nearer end of its own pass within 20 mm, over that pass's metal only, less the bead's own last 1 mm at a bead end (D1.1:2000 Table 6.1(3): "All craters shall be filled to provide the specified weld size"); everything solid and frozen in the pool (weld metal with no freeze time fails). Readings on cell edges are compared with a 10^{-9} m tolerance. No upper size bound and no leg-difference rule (D1.1 accepts an oversized fillet, and unequal legs when the smaller meets the size)	AWS D1.1:2000 (read), ISO 5817:2023 (read, authored use); end allowance, crater extent, start exemption and resolution authored

What the sections resolve. The weld's shape (legs, size, throat, convexity) is measured on the pool's free surface, the continuous profile each slice froze with, not on the cells. A cell is weld metal when its centre lies under that surface, so a leg read on the 0.25 mm cells would come out up to a cell short, and how short would depend on the face: a flat face would read ≥ 6.35 mm only from 6.50-6.55 mm. Each section's cells are checked against the surface. Metal the surface does not explain (droplets stacked or spilled after, metal frozen beyond a surface that drew back) is kept as whole cells, cells it claims that are not metal are removed, and beyond the toes the cells are kept as they are. Legs are the toes (the pool's 0.05 mm toe sampling). Size, throat and convexity are exact on the surface where the cells agree with it, else read on a 0.05 mm raster of both. On synthetic sections built as the pool builds them (true legs 6.30-6.60 mm, flat, convex and concave faces) the legs read exactly and the sizes within 0.0004 mm. A section with no surface (the cold start's stacked droplets) is read on the cells, and the leg-size check counts them. Fusion, penetration, overlap, undercut and porosity are read on the cells. Overlap is judged at one cell (0.25 mm): weld metal anywhere over parent metal that never fused beyond the fusion boundary, lying on it or standing over it (D1.1 Fig 5.4, weld metal protruding beyond the toe), per column of the base plate and per row of the bracket's face, so a face that overhangs its toe, with an air wedge under it, counts (`sections.tee_section`; a contract test builds both). Surfaces that overhang are read through the pool's own inside test (`pool.under_surface`) and polyline, as the cells were made.

The tack task's hand-held bracket is a design choice. Shops usually clamp a bracket against a stop or square before tacking; here the fixture's toggle clamps hold the base plate only and the FR3's hand holds the

bracket square while the UR10e tacks, so the task exercises two-arm coordination (the hand's grip, E19; the bracket moves 0.54 mm when the hand closes). A clamped single-arm variant is not built (§18).

Success is all of a task's criteria. Every task scores 1 on success and otherwise 0.9 × its progress; `fillet-2f`'s progress is the share of the required fillet that exists as sound, fused weld (below), and the other tasks' are in the eight-task table. Criteria that judge how a result was made count only when the result exists: the fillet's convexity, undercut, porosity and "solidified" only when a weld covers ≥ 90 % of the inspected seam (the share the leg-size rule asks to be full size; the overlap check also), the tack's squareness, position, released hand and cool joint only when the bracket is bonded; the torch's angles, no-contact, no-arc and settled checks only when the wire tip is within 5 mm of the root at the seam's start (the robots' home pose holds the torch at the procedure's angles, so ungated angle checks would pass on an untouched start state). Ungated, those criteria would pass vacuously and reward a policy for standing still; untouched start states score 0 (E22). Instructions are work orders in the trade's words; thresholds are in this report, not in the instruction. `fillet-2f`'s instruction gives the procedure's source settings (wire, gas, wire feed, voltage) and leaves travel speed, aim, angles and crater dwell to the policy: making the specified size at the set wire feed is the welder's part of the job, and in this model the procedure sheet's 14 in/min lays the fillet 2-3 % undersize (§18), so stating it would steer a policy into a failing weld; the contact tip to work distance is the torch frame's 19 mm (§9).

How fillet-2f is judged in detail (`task.py`). The instruction gives the procedure's source settings only. Two recorded runs that saved their section tables with freeze times serve as checks below: P1, the `aim_tacks` program replayed (SCRIPTED ROBOT), and P2, two runs meeting in the joint (ASSISTED phantom torch).

- *Craters.* Every place a pool froze last is a crater, wherever the weld stopped: a section with no later-frozen section of its own pass within 5 mm. A pass is a freeze episode, the sections inside the joint whose adjacent freeze times differ by ≤ 2 s; a section beyond a bracket end never splits one. Inside the joint the largest step within one pass is 0.79 s (P1, at the first tack's start), and a strike's first section freezes 3.7-3.8 s after the strike, so a restart struck at an earlier front meets it about 4 s apart and starts a new pass. A restart whose metal froze within 2 s of the earlier metal still merges. Each crater is judged ≥ 6.35 mm over 10 mm centred on it, and the window runs on from the stop to the nearer end of its own pass within 20 mm, a bead end or older metal, so a fall-off past the stop is judged. The window judges only its own pass's metal, so a later pass that stops near an earlier, unfilled crater without remelting it does not hide it. The bead's 1 mm end face is left out only at a bead end. A section at the edge of the sections read (151.75 mm), with freeze times rising toward it, is not a stop: the pass ran on beyond what was read. A size-0 hole inside a bead does not end it. A stop radius of 1 mm would read a step in the freeze times as a stop where `aim_tacks`' aim zone begins (6.25-6.75 mm); from 2 mm up only the stops are found. In `aim_tacks` the last-frozen sections are at 146.25-146.75 mm, under the arc's 1.2 s dwell (147 mm), and the crater judged is 141.25-149.75 mm (minimum 6.73 mm). In P2 the second run's window runs to 82.75 mm (minimum 6.49 mm). Residual: a later pool whose metal reaches an earlier stop from the side that pass came from, touching or remelting its last-frozen section, leaves the rest of that crater, toward the pass's front, judged only by the 10 % undersize rule. It is a band about 5 mm wide, not one exact position.
- *Ends.* Undercut and porosity are judged on the whole joint, overlap and convexity over the span and every crater. Within 5 mm of an end and outside every crater, a start's overlap is accepted up to two cells (0.5 mm), because every saved start leaves cold-lapped, stacked metal there (§18, cold starts); convexity is judged everywhere. The leniency is keyed to place: a start away from a joint end, as a middle-outward weld's, is held to one cell, and so is a lap running past 5.0 mm. `aim_tacks`' lap ends one slice inside the zone; keying the leniency to each pass's metal edge would take in the film beyond the bracket's end and fail `aim_tacks`. The overlap left unjudged is reported in each evaluation.
- *Cell edges.* Penetration, undercut and overlap are cell edges carrying float residue (one cell into the base plate reads 0.2499999999999933 mm), so they, the 5 mm end allowance and the 10 % undersize share are compared with a 10^{-9} m tolerance.
- *Acceptance basis.* D1.1 throughout. There is no leg-difference rule: D1.1 accepts unequal legs when the smaller meets the size, while ASME QW-184's rule would fail a 6.6 × 10 mm fillet and pass a 12 × 12 mm one. The porosity figure is an authored use of ISO 5817 C. The leg-size check reports the vertical leg's 0.25 mm steps.
- *Timing.* Weld metal whose slice the pool has not frozen fails `solidified`. The on-demand evaluation judges a weld still freezing with the arc off as it freezes hands-off, on a copy of the weld process; the world is not stepped. From the saved states of P1 and P2 that copy reproduces the run's own cooling in every section field (P1: 0.1 s simulated, 3.7 s wall; P2: 1.45 s, 39 s wall). The `aim_tacks` weld has 25 sections liquid at the end of its 0.5 s post-flow; its sections are solid 1.5 s after arc off, and its pool's last slices freeze 1.8 s after. The per-state evaluation (`state_evaluation`) judges the present state, and a liquid section earns no progress there.
- *Score.* 1 on success, else 0.9 × progress:
- progress = $(P - P_0)/(1 - P_0)$, clipped to [0, 1];
- P is the length-weighted mean over 5-145 mm of each section's credit: min(1, size/6.35 mm) if the section is solid, fused and free of overlap, undercut over 1 mm beyond the 50 mm the code allows, undercut over 2

mm, porosity and excess convexity, else 0;

- what success judges outside the 5-145 mm span costs progress: a crater section there costs its size deficit, and start-zone metal that is rejected costs all of it;
- $P_0 = 0.128$ is the tacked start's own value, so idle scores 0.

Nothing about direction, order, run count or schedule enters it. The mean of the 11 checks would rank a weld porous along its whole length (`no_gas` 0.545, `draft_4ms` 0.636) above sound, undersized ones (`fast_travel` 0.455), give a sound weld over three quarters of the joint 0.182, and fall when more weld joins the second tack; each evaluation still reports it as `reward.mean_of_checks`. `reward.net_progress` reports the unclipped value, and `judged` reports the time against `TIME_LIMIT_S`. The per-state progress reads 0 for the first seconds of a correct pass, while the first metal is still liquid or freezing, and lags the torch by the pool's freezing time; the final score is not affected. During P2's second pass it rose from 0 to 0.997, measured every simulated second, and dipped once, 0.806 to 0.801, while the pool remelted the second tack's credited metal, because liquid earns nothing.

Judged without physics on the saved rows: P1 succeeds, 1.0; P2 fails overlap only, 0.848, because its starts lap beyond two cells; the start state scores 0. The 20 recorded fillets of §14, without the start-zone charge (which lowers these scores by 0.01-0.03 and changes no verdict):

- `aim_tacks` 1.000;
- the 4.0-5.2 mm/s runs at 29 V aimed at the root 0.848-0.863, `travel_5p2_v31` 0.895, `travel_5p2_aim05` 0.832;
- Lincoln's 5.93 mm/s runs with gas 0.821-0.831;
- `fast_travel` 0.535 and `aim_up_2mm` 0.365;
- `no_gas`, `draft_4ms` and `long_ctwd` 0 (rejectable along their length).

`travel_5p0_x0_1`'s start is the only start besides `aim_tacks`' within the two-cell bound. The §14 table gives each run's verdicts as its evaluator judged them when it was recorded. Judged by the crater rule above, from the saved section tables (which keep no freeze times, so the stop is placed where the arc last stood, as P1 measured):

- the crater check passes `travel_4p8`, `travel_5p0_x0_1`, `travel_5p2`, `g1_travel_5p2`, `nominal`, `no_post_flow`, `dual_fr3` and `draft_4ms`. Their low sections are either the bead's last 1 mm (5.78-6.01 mm), or the 6.25 mm undersize 7-9 mm behind the stop, which the leg-size check judges;
- it fails `dual_panda` and `dual_panda_travel_5p2` (6.00 mm), `g1` (6.20), `travel_5p2_aim05` (6.23), `travel_5p2_v31` (5.33), `no_gas`, `fast_travel`, `aim_up_2mm` and `long_ctwd`;
- the leg-difference failures of `g1_travel_5p2`, `long_ctwd`, `aim_up_2mm` and `travel_5p2_aim05` in the §14 table do not apply, since there is no such rule.

No recorded success changes.

The coupon's placement varies by the frozen catalogue `welding-steel-cell-v1` (`welding/data/variations_catalogue.jsonl`): 10 000 entries with train / validation / test splits (7 982 / 983 / 1 035), each a coupon offset (±3 mm), yaw (±2°), the wire's stick-out left in the gun (6.35-12.7 mm) and a draft under the AWS limit (at most 0.6 % air in the shield at 2 m/s, under the porosity onset at the procedure's stand-off and angles); variation 0 is the authored layout. `gar run welding --variation N`, `--select split=test`.

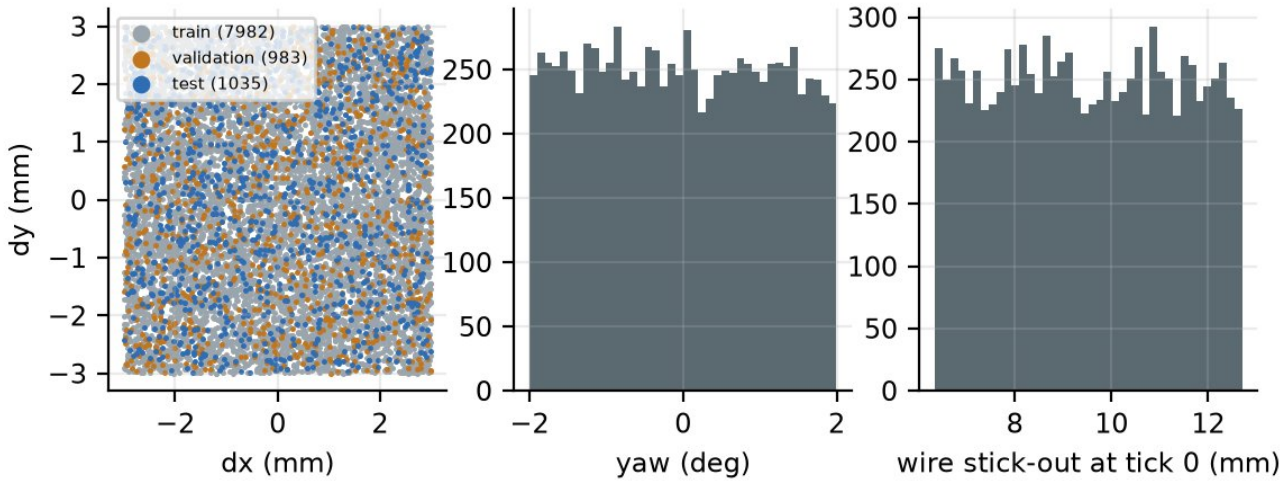


Figure 13.1 — The frozen catalogue: coupon offsets coloured by split, yaw and the wire's stick-out at tick 0.

The eight-task set

The benchmark's set is eight tasks: `task.py` (`EIGHT_TASKS`, the instructions, the verifiers and progress terms), `presets.py` (the three start presets), `variants.py` (which deployments offer which task) and `benchmark.json` (`task_sets`; eleven task ids × 10 000 variations = 110 000). The new tasks were not run in the simulator. Their verifiers were checked three ways:

- on constructed sections;
- on the section tables of P1 and P2, saved with their freeze times (§13 above);
- in the contract tests.

The set changes no physics, material, thermal, solver or robot parameter, so its physics and the cached start states are those of the rest of this report.

Score. Every task: 1 on success, else 0.9 × progress (fillet-2f's factor, `FAIL_SCORE_SCALE`), progress 0 at the untouched start state, continuous in the end state and blind to the route. The fraction of checks passed is reported as `reward.mean_of_checks`.

Which tasks are in the set. A task that cannot be shown to succeed as built is replaced by a solvable one; no standard is relaxed. Registered tasks outside the set stay in the catalogue, unchanged.

- `tack-and-weld` replaces `tack`. No demonstration passes tack's D1.1 checks (every robot tack fails `tack_fusion` and `tack_profile`; E26's nine starts fail). `tack` keeps its id, start, instruction and checks; its score follows the contract above.
- `half-fillet` replaces a crater-filling task. That task's start (a full fillet with a 0.5 s crater dwell) exists only after a scripted weld is run, and the crater fill restarts inside the span, where overlap is held to one cell (0.25 mm), while every start on record laps 0.5 mm or more (P1 0.5 mm, P2 3.0 and 3.5 mm, E26 ≥ 0.75 mm).
- `set-up-and-weld` replaces an interrupted-fillet task and `fillet-from-both-ends`. The interrupted fillet's start (P2's run A) exists only after a weld is run, and it carries a start lap of 3.5 mm at 149.75 mm and 1.0 mm at 149.25 mm, over the 0.5 mm start-zone bound. `fillet-from-both-ends` needs a start at the far end within the 0.5 mm start-zone lap, and none is on record (both of P2's starts lap beyond it: 3.0 mm at 0.25 mm, 3.5 mm at 149.75 mm; `aim_tacks`' 0.5 mm start is at the near end only). It stays registered outside the set.
- `hold-and-present` replaces `stitch-weld-ends`. With the only start on record (`aim_tacks`: 5.5 mm under size over 5.25-10.25 mm) the right 2 in window is 87.8 % full size against D1.1's 90 % (P1 to 58 mm with P2's run A beyond 92 mm, CONSTRUCTED). The 10 % rule is not relaxed. `stitch-weld-ends` stays registered outside the set.
- Considered and not taken: a weld-inspection or measurement task (the policy has no channel to report a measurement; an end state that encodes one, the wire on the root, is position-torch); a pre-weld cleaning step (the model has no surface contamination: `steel_mill_scale` is a render material); a gas or ground set-up step (a purge leaves no end state, the gas valve's end state is judged in set-up-gun; the ground clamp's 490 N magnet holds it).

#	Task	Difficulty	Start state	Success (all)	Progress
1	`position-torch`	entry	tacked	the §13 table	reach x pose; reach = 1 - clip((d - 1 mm)/(d0 - 1 mm)), d the wire tip's distance from the root line's 0-10 mm, d0 the start's; pose = mean of work angle (40-50°), push (0-15°), CTWD (17-21 mm), TCP motion < 0.5 mm, each 1 in band, 0 at twice its half-width outside; x 0.5 for gun contact or an arc in the episode
2	`set-up-gun`	entry	tacked; preset: source at 6.4 m/min, 20 V (`sc_0p9_C25_175A`), 6.35 mm of wire (a variation's stick-out replaces it)	source set points 12.7 ± 0.3 m/min and 29.0 ± 0.5 V (the source's state); stick-out 12.7 ± 1.5 mm (contact tip to work 19 mm - ¼ in, Lincoln L-56); no arc; no gun-hardware contact in the episode; gas valve closed	mean of the three terms, each 1 in band, 0 at the start's error, linear between, a term already in band at the start left out; x 0.5 for an arc or gun contact
3	`fit-up-bracket`	medium	loose; preset: bracket shifted 15-30 mm toward the weld side and yawed 3-8° about its vertical axis (seed 0: 25 mm, 6°), settled to rest	both ends of the bracket's root edge within 1.0 mm (`FIT_MM`) of the layout; squareness (tilt) <= 1°; hand not touching; bracket speed < 1 mm/s; base plate moved <= 0.5 mm; no arm latched in a protective stop	root-edge term net of the start (1 within 1 mm) x squareness (1 within 1°, 0 at 10°); 0 while the hand touches the bracket

#	Task	Difficulty	Start state	Success (all)	Progress
4	`half-fillet`	medium	tacked	over 5-75 mm: continuous fused metal, size >= 6.35 mm on 90 % and >= 3.95 mm everywhere, median throat >= 4.49 mm, every section sound (`_section_defects`, one cell of overlap), macro sections fused at 6, 37.5, 70 mm; beyond: judged craters' metal at one cell of overlap, the start zone at 0.5 mm, the episode's other metal free of convexity and porosity; every crater touching the half or of the episode's metal >= 6.35 mm (the second tack's, untouched, is the fixture's); undercut on the whole joint (<= 2 mm, > 1 mm on <= 50 mm); no metal in 100-120 mm; solid and frozen	raw = the half's sound-fillet credit (`section_credit`) less, per length, the judged craters' deficit beyond it and rejected start-zone metal; (raw - raw0)/(1 - raw0) - 0.5 x share of 100-120 mm holding metal, clipped
5	`hold-and-present`	medium	loose (the tack task's start); ur10e-fr3 only among the two-arm rigs (`variants.NOT_SHOWN`)	position-torch's eight checks; both Franka fingers touching the bracket; square <= 1° and centre offset <= 1.0 mm (`env.bracket_alignment`, the tack task's limits); no arm latched in a protective stop	0.5 x position-torch's progress + 0.5 x hold, hold = (both fingers on) x min(squareness, position terms: 1 in limit, 0 at twice)
6	`fillet-2f`	hard	tacked	the §13 table	`fillet_progress` (§13)
7	`set-up-and-weld`	hard	tacked; preset: source at 6.4 m/min, 20 V (`sc_0p9_C25_175A`), the wire as the start leaves it	every fillet-2f check; the source at 12.7 ± 0.3 m/min and 29.0 ± 0.5 V at the end	fillet_progress net of the tacked start (fillet-2f's)
8	`tack-and-weld`	hardest	loose	every fillet-2f check on the final weld (the tacks incorporated in it); bond active; square <= 1°; centre offset <= 1.0 mm; hand released	fillet_progress (net of the loose start: 0) x min(squareness, position terms: 1 in limit, 0 at twice) x 0.5 while the hand touches the bracket

Outside the set, registered: `tack`, `stitch-weld-ends` (windows 5-50 and 100-145 mm; middle 65-85 mm bare; craters judged only inside the windows) and `fillet-from-both-ends` (fillet-2f's checks and two runs from the ends meeting in 50-100 mm, read from the order the sections froze).

Instructions (`task.INSTRUCTIONS`, exact):

- set-up-gun: "Set the cell up for the spray procedure before welding: set the power source to the procedure's 500 in/min (12.7 m/min) wire feed and 29 V, and leave 1/2 inch of wire sticking out of the contact tip. Do not strike an arc, keep the gun clear of the parts and leave the shielding gas off."
- fit-up-bracket: "The bracket has been set down on the base plate off its layout. Fit it up for tacking: stand it upright and square to the plate, centred on the plate along and across it with its faces parallel to the plate's long edges, to within 1 mm, then let go of it." The layout is the plate's centre: plate and bracket share their centre in x and y (`process.tee_coupon`).
- half-fillet: "Weld a 1/4 inch fillet along the right half of the near side of the tacked bracket in the horizontal position: from the right end of the joint (the end on the robots' right, the right end in the bench camera's image) to its middle, 3 inches, and fill the crater where you stop. Leave the left half of the joint unwelded." + fillet-2f's spray schedule sentence.
- hold-and-present: "Get the joint ready for the first tack. With the hand, grip the bracket by its top edge and hold it upright and square on its layout; with the welding torch, come to the start of the near-side seam (the end on the robots' right, which is the right end in the bench camera's image) ready to weld: wire pointing at the root of the joint, 45 degree work angle, 5 to 10 degree push angle, 3/4 inch (19 mm) contact tip to work. Do not strike an arc; the nozzle must not touch the parts (touching them with the wire to find the seam is fine). Finish with the hand still holding the bracket and the torch in place."
- set-up-and-weld: "The power source was left on another job's settings. Set it to this joint's spray procedure, 500 in/min (12.7 m/min) wire feed and 29 V for 0.035 inch ER70S-6 wire with 90/10 gas, then weld a continuous 1/4 inch fillet along the near side of the tacked bracket in the horizontal position, from end to end, and fill the crater." The policy's feedback reports the source's actual feed and voltage, 0 with the output off, not its set points, so the start condition is stated.
- tack-and-weld: "The bracket stands loose on its layout on the clamped base plate. Tack it square to the plate at both ends of the near side, holding it in place while you tack, then let go and weld a continuous 1/4 inch fillet along the near side in the horizontal position, from end to end, and fill the crater." + the schedule.

The schedule is quoted because the posted procedure sheet (the board behind the robots, `shop.py`) is not shown to be legible in the policy cameras. Time limits (simulated): set-up-gun 60 s, fit-up-bracket 120 s, half-fillet 120 s, hold-and-present 90 s, fillet-2f 180 s, set-up-and-weld 180 s, tack-and-weld 300 s.

Checks on recorded rows (pure evaluation of saved section tables; CONSTRUCTED where marked):

- Every weld task scores 0 on its start rows (the tacked start of P1/P2; a bare joint for tack-and-weld). set-up-gun and fit-up-bracket score 0 on a stub world at their preset starts (CONSTRUCTED).
- set-up-and-weld: P1's final rows with the source at the procedure pass (score 1.0, progress 0.997). With the source left at 6.4 m/min / 20 V they fail `source_at_procedure` only (score 0.898).
- half-fillet:
- P1's full fillet fails `left_half_unwelded` only (score 0.446).
- P1's rows cut at 80 mm onto the tacked start (CONSTRUCTED) pass.
- P2's run A, an ASSISTED pass stopped in the joint at 75 mm with a 1.2 s dwell, leaves a crater judged over 73.75-83.25 mm at a minimum of 6.49 mm, so a crater filled where a weld stops in the joint passes this rule.
- tack-and-weld: P1's final rows with an assembled bracket (bond, 0.15°, 0.3 mm, released: CONSTRUCTED values) pass.
- Outside the set:
- stitch-weld-ends: P1 to 58 mm with P2's run A beyond 92 mm (CONSTRUCTED) fails `leg_size_right_end` only.
- fillet-from-both-ends: P2's final rows fail `overlap` only (score 0.852); P1 fails the run check only (score 0.590).

Solvability basis and open risks:

- position-torch: SCRIPTED ROBOT on three rigs (ur10e-fr3, dual-panda, g1), each passing all eight checks. No material risk.
- set-up-gun: the `tool_io` set points and inching are contract-tested; E6 inched the wire 11.5 -> 20 mm (§11.3).
- Risk: the policy cannot read the stick-out (its `tool` feedback has no stick-out), and whether 1-2 mm is resolvable in the cameras is not shown. Touch sensing against the plate at a known stand-off measures it (E6 case b, within 0.35 mm).
- 2,333 of the 9,999 variations with a stick-out (23 %) already hold 11.2-12.7 mm of wire: their stick-out term is left out.
- fit-up-bracket: E19 grip; E24 (b) lifted the loose bracket 20 mm and set it back 0.20 mm off its place.
- Risks: a carry of 15-30 mm with a 3-8° turn is not shown.
- The displaced start is built by `presets._place_free_body` and settled by the start states' rest gate. That gate fails the build loudly if the bracket does not come to rest or a protective stop fires, but the build has not been run: the robots' home poses are not checked clear of the displaced bracket.
- The 1 mm limit on both root ends allows about 0.76° of yaw.
- half-fillet: the `aim_tacks` start and travel cover 0-75 mm. That is one SCRIPTED ROBOT run passing all 11 fillet checks (§14), which P1 replayed exactly; its 5.5 mm of start undersize is within the 7 mm the half allows. The crater filled where the weld stops in the joint is shown by an ASSISTED run (P2's run A). The half has not been run as one robot program.
- hold-and-present: each interaction is shown by a SCRIPTED ROBOT demonstration on ur10e-fr3.
- The torch's presentation: position-torch, wire inched to 19 mm and touch-sensed onto the root at x = 1 mm.
- The hold, with the torch at the seam's near end: the tack demonstration. The hand grips the loose bracket's top edge at x = 120 mm with 30 N per finger, and the torch is brought to the first tack's start (45° work, 5° push, 19 mm contact tip to work) while it holds. Its square and position checks pass at the end.
- The grip's own disturbance: E24 (b), 0.058° and 0.068 mm after the grip.
- The arc struck on this loose start (the tack demonstration), so its circuit closes.
- Not run as one program: presentation at x = 1 mm on the loose start while the hand holds.
- On dual-fr3 and dual-panda the task is refused (`variants.NOT_SHOWN`): the arms met with the torch at the near end and the hand at x = 120 mm, and their demonstrated order holds the near end.
- set-up-and-weld: the recorded `aim_tacks` program commands the procedure's set points itself before it strikes (`scripted.WeldProgram.weld`: `tool_io(gas_on, wire_feed_m_min=12.7, voltage_v=29)`, then `arc_on`). With the output off the source's set points enter nothing the process computes (`arc.CVSource.update`: the wire inches at `RUN_IN`). So from this start, which differs from the tacked start only in the set points, the program runs as recorded, and that run passes every fillet-2f check (P1). No material risk beyond fillet-2f's own (one passing program).
- tack-and-weld: the tacks (SCRIPTED ROBOT on three rigs: bond, square 0.12-0.20°, released) and the fillet over tacks (`aim_tacks`, over the fixture's tacks) are each shown.
- Risk: the fillet over robot tacks (6.45-6.5 mm, one to three convex sections) is not shown.
- Its success judges the end state: a bracket held and welded without tacks is judged the same.

14. Assisted demonstrations

Every demonstration is labelled in its film's caption band and in its result file. **SCRIPTED ROBOT**: the robots act only through the policy's commands (joint targets from IK, the Franka Hand, `tool_io`); the assistance is that the program is written from the coupon's known pose instead of being found by a policy. **ASSISTED**: a phantom torch or a phantom force that no policy can produce (the tack fixture, E4, E7). **UNASSISTED**: nothing is commanded. The unmodified task evaluator judges every task demonstration; nothing in it reads the program.

The demonstrations (each judged by the unmodified task evaluator when it was recorded, with D1.1's size, crater and tack checks; §13 gives the crater verdicts under the crater rule described there):

Demo	Deployment	Task	Procedure	Evaluator	Fails	Measures
aim_tacks	ur10e-fr3	fillet-2f	5 mm/s, 29 V, start x 1 mm	11 of 11 (success)	-	>= 6.35 mm on 93 %; overlap 0.25 mm over 0.0 mm
aim_up_2mm	ur10e-fr3	fillet-2f	5.93 mm/s, 29 V, aim +2 mm	2 of 11	continuous_length, leg_size, leg_difference, throat, convexity, undercut, overlap, fusion_sections, crater_filled	>= 6.35 mm on 1 %; overlap 8.25 mm over 41.5 mm
crater_1p2	ur10e-fr3	fillet-2f	5.93 mm/s, 29 V	9 of 11	leg_size, overlap	>= 6.35 mm on 19 %; overlap 1.25 mm over 8.0 mm
draft_4ms	ur10e-fr3	fillet-2f	5.93 mm/s, 29 V, draft 4 m/s	7 of 11	leg_size, overlap, porosity, crater_filled	>= 6.35 mm on 18 %; overlap 1.25 mm over 8.0 mm
dual_fr3	dual-fr3	fillet-2f	5.93 mm/s, 29 V	8 of 11	leg_size, overlap, crater_filled	>= 6.35 mm on 23 %; overlap 1.50 mm over 8.0 mm
dual_panda	dual-panda	fillet-2f	5.93 mm/s, 29 V	8 of 11	leg_size, overlap, crater_filled	>= 6.35 mm on 22 %; overlap 1.25 mm over 7.5 mm
dual_panda_travel_5p2	dual-panda	fillet-2f	5.2 mm/s, 29 V	9 of 11	overlap, crater_filled	>= 6.35 mm on 98 %; overlap 0.75 mm over 6.0 mm
fast_travel	ur10e-fr3	fillet-2f	9 mm/s, 29 V	5 of 11	continuous_length, leg_size, throat, convexity, overlap, crater_filled	>= 6.35 mm on 0 %; overlap 2.25 mm over 13.5 mm
g1	g1	fillet-2f	5.93 mm/s, 29 V	8 of 11	leg_size, overlap, crater_filled	>= 6.35 mm on 20 %; overlap 1.25 mm over 8.0 mm
g1_travel_5p2	g1	fillet-2f	5.2 mm/s, 29 V	7 of 11	leg_size, leg_difference, overlap, crater_filled	>= 6.35 mm on 97 %; overlap 0.75 mm over 7.0 mm
long_ctwd	ur10e-fr3	fillet-2f	5.93 mm/s, 29 V, CTWD 30 mm	5 of 11	continuous_length, leg_size, leg_difference, convexity, overlap, crater_filled	>= 6.35 mm on 0 %; overlap 7.25 mm over 73.0 mm
no_gas	ur10e-fr3	fillet-2f	5.93 mm/s, 29 V, no gas	6 of 11	leg_size, throat, overlap, porosity, crater_filled	>= 6.35 mm on 16 %; overlap 0.50 mm over 4.0 mm

Demo	Deployment	Task	Procedure	Evaluator	Fails	Measures
no_post_flow	ur10e-fr3	fillet-2f	5.93 mm/s, 29 V, no post-flow	8 of 11	leg_size, overlap, crater_filled	>= 6.35 mm on 18 %; overlap 1.25 mm over 8.0 mm
nominal	ur10e-fr3	fillet-2f	5.93 mm/s, 29 V	8 of 11	leg_size, overlap, crater_filled	>= 6.35 mm on 18 %; overlap 1.25 mm over 8.0 mm
position-torch	ur10e-fr3	position-torch	-	8 of 8 (success)	-	wire 0.40 mm from the root
position-torch_dual-panda	dual-panda	position-torch	-	8 of 8 (success)	-	wire 0.39 mm from the root
position-torch_g1	g1	position-torch	-	8 of 8 (success)	-	wire 0.83 mm from the root
slow_travel	ur10e-fr3	fillet-2f	4 mm/s, 29 V	10 of 11	overlap	>= 6.35 mm on 100 %; overlap 0.50 mm over 5.0 mm
tack	ur10e-fr3	tack	-	7 of 9	tack_fusion, tack_profile	bond 67.2 kN
tack_dual-fr3	dual-fr3	tack	-	7 of 9	tack_fusion, tack_profile	bond 67.1 kN
tack_dual-panda	dual-panda	tack	-	7 of 9	tack_fusion, tack_profile	bond 64.7 kN
travel_4p8	ur10e-fr3	fillet-2f	4.8 mm/s, 29 V	9 of 11	overlap, crater_filled	>= 6.35 mm on 99 %; overlap 0.75 mm over 6.5 mm
travel_5p0_x0_1	ur10e-fr3	fillet-2f	5 mm/s, 29 V, start x 1 mm	9 of 11	overlap, crater_filled	>= 6.35 mm on 100 %; overlap 0.50 mm over 5.0 mm
travel_5p2	ur10e-fr3	fillet-2f	5.2 mm/s, 29 V	9 of 11	overlap, crater_filled	>= 6.35 mm on 98 %; overlap 0.75 mm over 5.5 mm
travel_5p2_aim05	ur10e-fr3	fillet-2f	5.2 mm/s, 29 V, aim +0.5 mm	7 of 11	leg_size, leg_difference, overlap, crater_filled	>= 6.35 mm on 89 %; overlap 4.00 mm over 9.0 mm
travel_5p2_v31	ur10e-fr3	fillet-2f	5.2 mm/s, 31 V	9 of 11	leg_size, crater_filled	>= 6.35 mm on 80 %

Every row records no protective stop and no source fault; the fillet rows' metal equals their deposit (within 0.8 mm³ of about 1000), their wire balance is 0 and their energy ledgers close to 10^{-8}. Every result file also reports weld metal with no face on steel (`sections.floating_weld`): none in 24 of the 27 demonstrations; 1 piece of 2 cells (0.06 mm³) in `dual_fr3`, 3 pieces (0.13 mm³) in `aim_up_2mm` and 5 single cells (0.16 mm³) in `fast_travel` (§18).

What the table says, in words:

- **A scripted program passes the fillet task from the start state every policy receives** (`aim_tacks`, 11 of 11, SCRIPTED ROBOT on the UR10e with the FR3): Lincoln's schedule at 5.0 mm/s from x = 1 mm with a 1.2 s crater dwell, the wire aimed 1 mm out onto the plate over each tack's start (x = 5-13 and 124-132 mm). The size reaches 6.35 mm on 93 % of the inspected length (median 6.71 mm), the crater 6.73 mm, the overlap at most one 0.25 mm cell. Aiming onto the plate where the tacks' cold laps lie is what a welder does to wash them in; the policy must find it. The same program without the aim (`travel_5p0_x0_1`) fails overlap (0.5 mm over 5 mm, over the tacks' starts) and the crater (5.78 mm with its 0.5 s dwell). E27 (below) shows the same on the weld process alone.
- **Lincoln's nominal procedure lays the fillet 2-3 % undersize**: a 6.25 mm median size on every robot (≥ 6.35 mm on 18-23 % of the length), the metal exactly the deposit (§18); 4.8-5.2 mm/s reach size on 97-100 %. The nominal runs pass 8 of 11 and fail size, overlap (1.25-1.5 mm over the first tack's start) and the

crater.

- **The crater is a technique, not a wall**: the 0.5 s dwell leaves 5.7-6.25 mm over the last 10 mm; a 1.2 s dwell fills it to 6.45 mm (`crater_1p2`).
- **The negative controls fail what they govern**: no gas (porosity, throat), a 4 m/s draft (2.6 % air, porosity 8 %), 9 mm/s (size, throat, continuity), a 30 mm stand-off (continuity, size), aiming 2 mm up the bracket (2 of 11), 4 mm/s (too much metal: 7.53 mm median size, overlap). `travel_5p2_v31` (5.2 mm/s at 31 V) fails size (≥ 6.35 mm on 80 %) and the crater.
- **The draft demonstration tests one thing.** At 4 m/s the air under the arc (2.6 %) stays far below the 8.9 % at which spray stops, so the arc, its current, spatter and deposit, and the weld's cells are the nominal run's: the same sections and the same failures besides porosity. It demonstrates the porosity bookkeeping, not a draft's effect on the arc or the bead.
- **The robots weld alike**: the dual-FR3, the dual-Panda and the G1 lay Lincoln's nominal fillet 8 of 11 with the same failures as the UR10e. The program is the same; only the arm that carries the gun changes.
- **The tack task** scores 7 of 9 on the UR10e with the FR3, the two FR3s and the two Pandas: bonded (65-67 kN), square (0.12-0.20° with the tacks' shrinkage), released and cool, but every robot's tacks fail D1.1's tack checks. Their starts are nearly sound (the robot program's back-step start: 0.5 mm unfused at each run's first section, overlap 0.5-2.0 mm, the largest a lip at the dual-Panda's first strike, x = 139.25 mm), but each tack has one to three sections over the convexity limit, and the two-arm rigs' tacks are 6.45-6.5 mm, over D1.1's ¼ in tack size (§13). The G1 is not offered the tack task (§9).
- **The position-torch task passes on every robot that ran it**, 8 of 8.

E26: a tack's start under each starting technique (ASSISTED: the weld process alone with a phantom torch on a fresh coupon, judged by the tack task's own checks). Nine techniques on the tack fixture's schedule; none passes every tack check. The start depends on the technique as in the shop: straight 4.5 mm of overlap and 1.5 mm unfused at the start; a soft start (6.4 m/min, 28 V) worse (6.25 mm, 3.5 mm unfused); a 0.4 s dwell 3.25 mm; 32 V hotter 2.0 mm; Fronius's 135 % start current 4.0 mm; the back-step starts the best (a back-step strike 8 mm inside: 0.75 mm overlap, 0.5 mm unfused, size 6.25 mm, two sections over the convexity limit; the robot program's, strike 12 mm inside: 0.75 / 0.5 mm, size 6.45 mm). No weld metal without a face on steel in any of them, and the metal is the deposit within 0.8 mm³.

E27: fillet programs over the start state every policy receives (ASSISTED: the weld process alone, restored from the tack fixture's cached start state, judged by the task's fillet evaluator). Three programs at Lincoln's schedule, 5.0 mm/s from x = 1 mm, 1.2 s crater dwell. **The program that aims 1 mm onto the plate over each tack's start passes every fillet-2f check** (overlap at most one 0.25 mm cell, crater 6.73 mm, size ≥ 6.35 mm on 94 %), as the robot does with it (`aim_tacks`). Without the aim the nominal program fails overlap only (0.5 mm over 4 mm, over the first tack's start; crater 6.73 mm, size on 100 %), and slowing to 3.5 mm/s over the tacks' starts instead fails overlap only (0.75 mm over 1 mm). No weld metal without a face on steel in any of them; the metal is the deposit within 0.74 mm³ of 4300-4500.

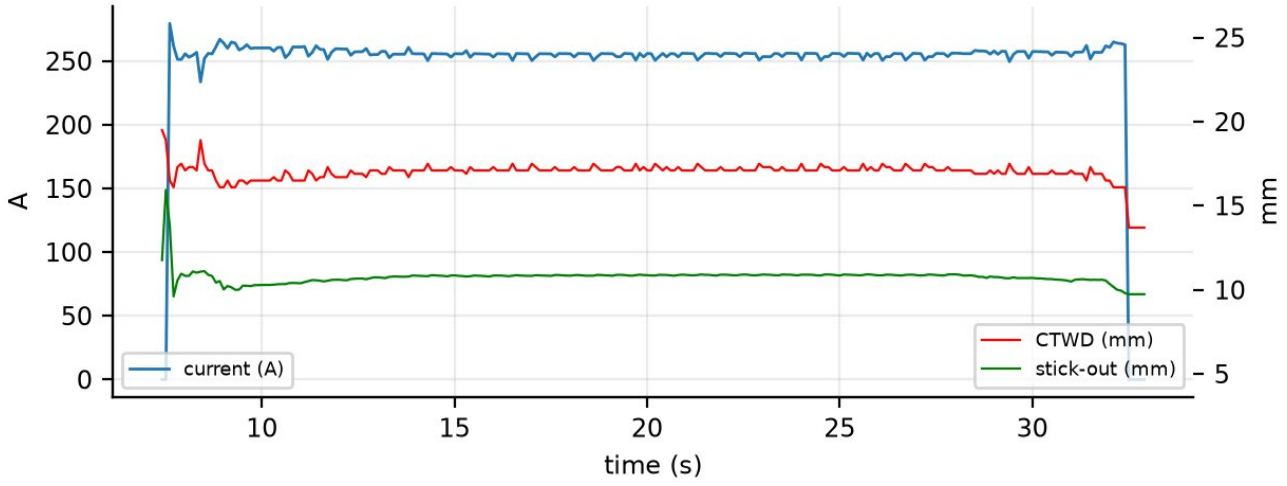


Figure 14.1 — The nominal robot weld: the arc current and the wire's stand-off and stick-out against time, sampled every 0.1 s.

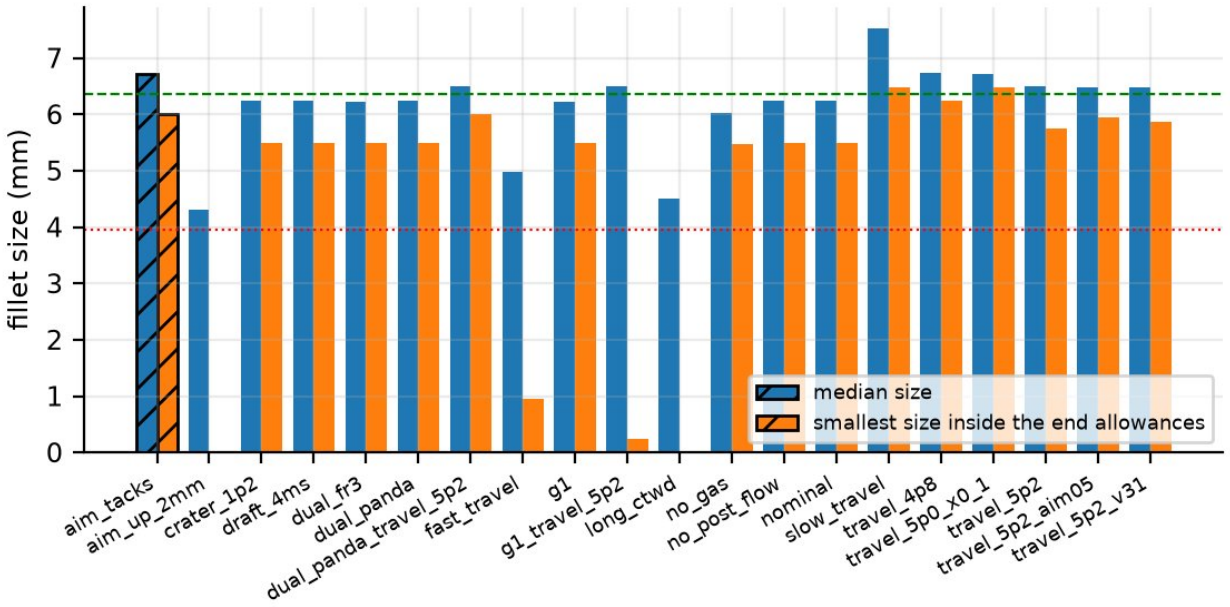


Figure 14.2 — Median fillet size of every fillet demonstration and the smallest size inside the end allowances (the evaluator's), successes hatched (green: the ¼ in nominal; dotted red: the D1.1 underrun floor).

15. Rest, soak, legal abuse and stress

At rest (E0, UNASSISTED). Every offered start state of every deployment is left for 20 s with nothing commanded. That is the loose bracket standing on the clamped plate (tack; the four deployments with a handling hand) and the tacked assembly with the bond active (fillet-2f, position-torch; all six). The check: after a 1 s settle, no free body (coupon, clamp pads, ground clamp), no torch work point and no hand tool point moves more than 0.05 mm, nothing is non-finite, no protective stop. Before handover the environment settles each start state hands-off until it is at rest (`env.settle_to_rest`: 0.3 s, then 1 s windows until nothing E0 gates moves 0.025 mm in one; handed over after a fixed 0.3 s, the G1 would still sway, its torch 25.7 mm peak to peak in the first second). The standing G1 is held 120 s, not 20 (`ROBOT_REST_S`), because a slow slide can hide in 10-30 s windows. **10 of 10 pass.** On the five fixed-base deployments the loose starts' worst point moved 0.07 µm, and on the tacked starts the bracket moved 3.96 µm over 20 s: the tacks' shrinkage still turning it a little as they cool their last degrees (§8, E25). The G1's first settle from its import pose takes 13.3 simulated s (22.6 s of wall time), once per physics version (cached; later starts settle 1.3 s). Over the 120 s after it its torch moved at most 0.0156 mm (tack start) and 0.0114 mm (fillet-2f start), its pelvis 0.34-0.48 µm after the first second: no steady drift. Wall time per simulated second: 1.6-1.7 for a loose start (cold coupon: the thermal solver is skipped exactly), 17-20 for a tacked start (the coupon still warm from its tacks), the G1 1.7 and 17.4. The rest films show 30 s of the tacked start from the cell's high camera and 120 s of the G1's.

Long soaks (E21). The default deployment's start states are each left 120 s with nothing commanded; then the nominal program welds x = 20-40 mm and the cell rests 60 s. The start states of the other five deployments are left 30 s each (`--robot-seconds`). Gates: E0's creep rule, measured from the 1 s point; no protective stop; finite state; no heating once no arc and no liquid remain; the weld solidified; every E20 check below. The G1's start states are held 120 s (`ROBOT_SECONDS`). **10 of 11 pass.** The largest drift of any free body over the start states: 0.42 µm (tack start), 5.5 µm (fillet start), 4.5 µm on every other fixed-base deployment, and the G1's torch 0.0154 mm (tack start) and 0.0114 mm (fillet start) over 120 s. **The weld-and-rest case fails creep**: after the weld the bracket moved 0.058 mm over the 60 s rest (gate 0.05 mm), the new fillet's shrinkage turning it as it cools (§8). That is the weld's own distortion, which the creep gate (written for a cell at rest) does not separate from drift; the gate was not changed. The metal after the weld matches its deposit to 2.5×10^{-5}. The base plate creeps at about 4 nm/s at rest (§18); the 0.05 mm gate is far above it.

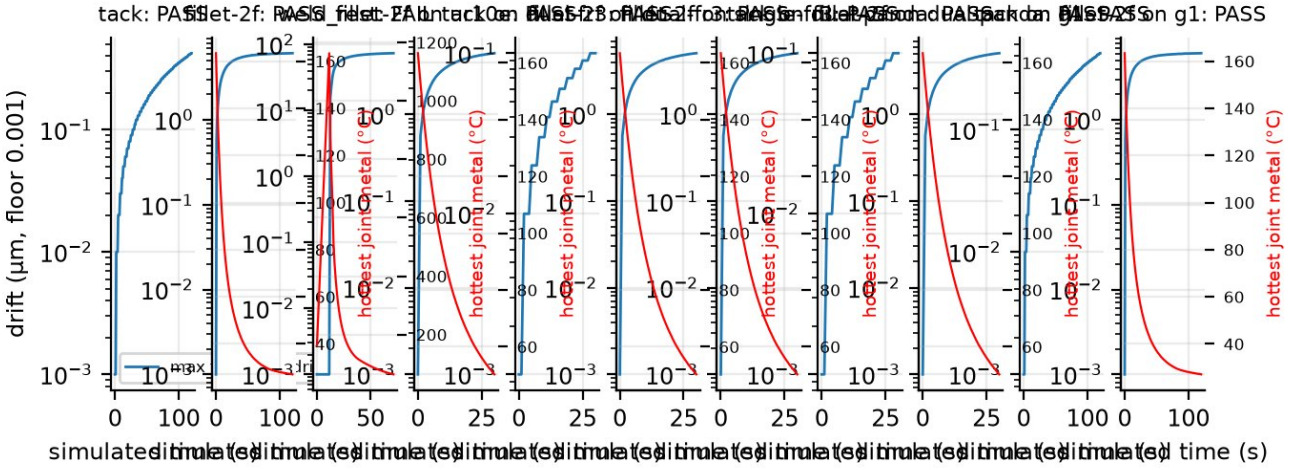


Figure 15.1 — E21: the largest free-body drift (µm) and the hottest joint metal over each soak.

Legal abuse (E8, SCRIPTED ROBOT). Commands a policy may legally send that a cell has to survive: (a) the torch driven 25 mm past the root at 20 mm/s; (b) the arc parked 8 s at one spot; (c) the arc switched on and off every 20 ms for 2 s while travelling; (d) the arc with the contact tip 8 mm from the work; (e) the source at its maximum, parked. **Three of five pass**; (d) and (e) fail the predictions declared for them before they were run, because the arc stubs: a parked pool stands as one bulge under the wire, the bulge reaches the wire and the wire drives into it. The predictions were not rewritten after the fact:

Case	What happened
(a) torch 25 mm past the root	the nozzle met the parts and the UR10e's protective stop fired at 150.0 N at the tool (its 150 N limit at the tool and the elbow, §9); finite
(b) arc parked 8 s	the pool grew into a 159 mm² section with 13 / 14 mm legs and 1.5 mm of overlap (x = 76 mm), fused 2.5-3.75 mm into the parts; hottest metal 2649 °C (below the 2860 °C boiling cap); evaporation 6×10^{-6} of the input; finite
(c) arc on/off every 20 ms for 2 s	15 strikes, each by run-in; no fault; metal 422.19 mm³ for 422.18 mm³ deposited; energy ledger closed to 3.9×10^{-8}

Case	What happened
(d) contact tip commanded 8 mm from the work	**FAIL (prediction: mean current above 280 A)**. The nozzle rested on the plate at 41-44 N (an 8 mm contact tip to work puts it on the joint). The pool built up under the parked wire, the stick-out burned back to 0.21 mm and the wire stubbed 0.16 s after the output went on (66 A mean); the stub's push lifted the gun off the plate, and when the arm's servo brought it back the nozzle met the plate at 423-493 N for one tick and the UR10e's 150 N force limit stopped the arm (every tick logged: a press against the plate, not a collider kick); finite
(e) the source's maximum, 25 m/min at 44 V, parked 5 s	**FAIL (prediction: no fault)**. The source held its 400 A ceiling until the wire stubbed into the pool 1.2 s after the strike (`stubbing`); the metal ends 0.29 % above the deposit, nothing spilled, evaporation 3.7 % of the input, ledger closed to 3.0×10^{-8}, the tack bond held; finite

Legal-action stress (E20; SCRIPTED ROBOT and RANDOM legal actions only). Every action goes through the welding backend's own validation (joint limits, IK, the hand's limits, `tool_io` names and ranges) and is applied as the backend applies it; nothing is written into the world. 16 pre-registered cases (random joint targets every tick, slams of the torch into the table, a clamp and the joint, the arms commanded into each other, the hand squeezing and yanking the loose and the tacked bracket, the wire inched into metal, touch sensing pressed into contact, every surroundings collider rammed 50 mm deep, the arc rammed, toggled every tick and parked 15 s, the source at its maximum and its minimum) and 12 seeded random episodes of every action kind, 5 % of them out of range (which the backend must refuse with no side effect). Checks every 50 ticks: no world fault, finite rigid and thermal state, bodies inside the cell, free-body speed under 10 m/s, coupon penetration under 1 mm, the energy ledger under 10^{-3}, metal against deposit within 10 %, temperatures between ambient and the boiling cap, an arc only through a closed circuit, the wire's stick-out within its bounds and never inside metal, a bonded bracket tilting under 1°, refusals leaving no trace, the evaluator running, and the world usable afterwards.

Two cases test the source's limits: the lowest wire feed at the highest voltage (1.3 m/min, 44 V) and a voltage far too low for the feed (12.7 m/min, 12 V), with a gate on the wire balance (fed − deposited − spatter − stick-out growth within 0.001 mm³). `long_arc_dwell` is held until the pool is solid (up to 60 s after the arc goes off, the hottest metal sampled every 50 ms) with three gates: no rise of the hottest metal with the arc off, frozen within a conduction estimate of its freezing time, and the frozen metal within 1 % of the deposit. Every case ends with `weld_metal_supported`: no weld metal without a face on steel (`sections.floating_weld`). Result: **28 of 30 cases pass**; 287.1 s simulated, 103 protective stops, 80 refused actions, 5655 s of wall time. The worst values over all cases: free-body speed 1.82 m/s, coupon penetration 0.46 mm, energy-ledger residual under 10^{-6}, the wire balance 0 in every check, a bonded bracket tilting 0.043°. The source's limits act in the world: at 1.3 m/min and 44 V the arc burns back after the strike and stays out (`burnback`); at 12 V the wire stubs at the strike; the source at its maximum while travelling held 400 A and then stubbed 0.12 s after the strike, the metal within 0.005 % of the deposit. **Two cases fail.**

- `long_arc_dwell` (the spray arc parked 15 s at one spot). The arc stubbed after 9.0 s: the bead, with 16 mm legs (a 226 mm² section at x = 103 mm), had grown up to the wire, and the nozzle touched it for 2.2 s. Before that the pool had outgrown the region the pool model may change (14 mm along each member), and up to 561 mm³ of its metal waited unplaced: the metal fell 505 mm³ behind the deposit (gate 10 %). After the arc went out the pool was solid 0.55 s later with no rise of the hottest metal (`cools_after_arc_off` passes), but the freezing-time gate had no estimate to check against (nothing was above liquid fraction 0.5 at arc-off), the frozen metal ended 37.9 mm³ (2.4 %) over the deposit (gate 1 %), and 15 pieces of weld metal (1.3 mm³) had no face on steel, the largest 13 cells at the pool region's upper edge (z = 15.6-15.9 mm). A 15 s parked spray arc is far outside any procedure; the region's size and what happens past it are the model's limits (§18). A 5 s park, run on the weld process alone, froze 3.35 s after arc-off with no rise and +0.5 % metal.
- `touch_sense_during_contact` (touch sensing pressed into contact, then a 0.3 s arc) fails only the `weld_metal_supported` gate: 3 single cells of weld metal (0.09 mm³) with no face on steel, one inside a groove in the bracket (§18).

Ramming the surroundings showed the colliders acting on an arm, each with a protective stop. After every case no arm was left latched in a stop. In some cases the probe's one command home did not get there within 2 s, because it arrived while the arm's 1 s stop was still latched, or because the way home ran through a collision that stopped it again; a separate check shows a stopped arm moving away once a command backs it off.

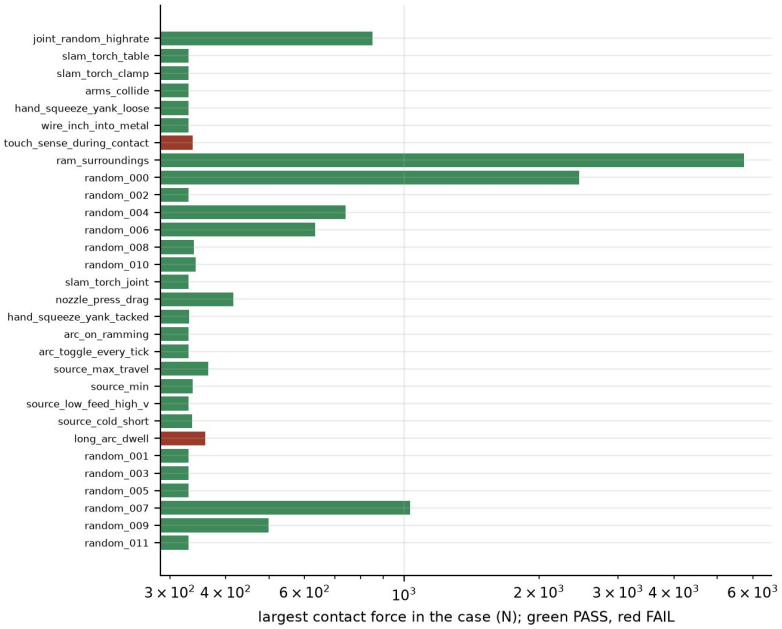


Figure 15.2 — E20: the largest contact force in each case (log scale); green PASS, red FAIL.

Catalogue entries at rest (E22: PASS 9 of 9). Nine pre-registered entries of the frozen catalogue (the first nine by a hash of their ids, not stratified: five train, one validation, three test), the tasks in rotation, built through the backend: the live coupon pose within 0.034 mm in plane, 0.14 mm in height (the clamped plate settling) and 0.0005° of the entry; 5 s at rest with no stop and the largest coupon drift 1.6 µm (the bracket, its tacks' shrinkage still turning it as they cool, §8); every untouched start state scoring 0. The nine entries took 178 s of wall time on the loaded GPU (the start states built and cached by then), every entry starting with the settle to rest (E0 above).

16. Runtime

Measured wall time per simulated second on this machine (RTX 5090 shared with other jobs, 16 CPU cores), from the result files' own timers (`wall_s / sim_s` of the whole episode).

Run	Wall s per simulated s
E16, `gar run` + HTTP, policy key, one camera (bench), arc-on travel commands each returning an observation: median per command length	500 ms 107, 200 ms 122, 100 ms 191, 50 ms 262, 20 ms 372, 10 ms 585; worst gated request 220 (batch: tool_io 1 tick + travel 100 ms #1 (arc on)); fixed cost per request 6.73 s + 94 × simulated
the same, observation only (no simulated time) / session launch	1.05 s (max 1.09 s) of wall time / 37.2 s
E16, `gar run` + HTTP, policy key, three cameras (bench, side, weld_camera), arc-on travel commands each returning an observation: median per command length	500 ms 140 (max 161), 200 ms 191, 100 ms 263, 50 ms 361, 20 ms 657, 10 ms 1121; worst gated request 272 (batch: tool_io 1 tick + travel 100 ms #1 (arc on)); fixed cost per request 12.07 s + 120 × simulated
the same, observation only (no simulated time) / session launch	5.50 s (max 5.54 s) of wall time / 48.8 s
E16, `gar run` + HTTP, policy key, no cameras (backend option `cameras: []`, no render worker), arc-on travel commands each returning an observation: median per command length	500 ms 133, 200 ms 143, 100 ms 167, 50 ms 177, 20 ms 210, 10 ms 172; worst gated request 177 (travel 50 ms #1 (arc on)); fixed cost per request 1.99 s + 131 × simulated
the same, observation only (no simulated time) / session launch	0.00 s (max 0.00 s) of wall time / 19.3 s
E0 at rest, loose start state (cold coupon: the thermal solver is skipped exactly), fixed-base deployments / the G1	1.6-1.7 / 1.7
E0 at rest, tacked start state (coupon still warm from its tacks: the thermal solver runs), fixed-base deployments / the G1	17.0-20.3 / 17.4
E21 soaks: loose / tacked start state at rest; a weld then rest	1.53 / 14.7; 39 then 13.7
robot welds without rendering, whole episodes (strike, travel, crater, post-flow, cooling; 12 runs on 3 deployments)	42-90 (the weld process 91-96 % of it; with the G1 92-94 %)
tack and position-torch demonstrations without rendering (4 runs)	20-25
E4 phantom-torch fillet (no robot, no rendering)	67
E12: 0.35 mm cells / reference / pool every 2 ms	47 / 65 / 114
E20, the slowest stress case (the spray arc parked 15 s)	118
films (a 1280 × 720 frame of a shop camera's view and two or three path-traced close-ups every 100-200 ms; 10 demonstration films)	51-121; **185.7 for the nominal weld's film** (four views every 100 ms: 330 frames at a median 10.9 s each; its physics alone 73)

Per-request cost and the shortest command (E16). A request through `gar run` + HTTP costs a fixed part plus the physics. The fixed part is almost all the observation's images: the frame of the physical state (about 0.3-0.5 s, most of it the joint band's surface) and one Isaac RTX render per requested camera, all cameras in one worker request: an unchanged scene re-renders in 0.06 s, any change costs about 0.2-0.6 s per camera (0.5-0.6 s with the arc burning, for the arc light, fume and pool reflection), any moved body about 0.5 s more. The rest is small: the state, read once per tick (0.04 s), the action journal (three SQLite commits, about 0.1 s, with spikes to 1 s on the shared disk) and the per-action trace (under 0.01 s). The task evaluator runs on demand (`GET /v1/evaluation`), not with every command.

E16 (run while the GPU carried eight other processes, at most one other Isaac render worker, 10.3-10.5 GiB in use and 100 % busy: medians of `nvidia-smi` every 3 s, recorded in the file for each camera set). E16 gates arc-on commands of 100 ms or longer when images are rendered and of 50 ms or longer without images, because the 50 ms commands' margin with images depends on the load. **With one camera** 100 ms commands did not all stay within 200×: 191× median, worst 220× (a 1-tick `tool_io` batched with a 100 ms travel); 500 ms ran at 107×. **With three cameras** 100 ms ran at 263× (worst 272×) and 500 ms at 140× (at most 161×). **Without images** (backend option `cameras: []`: no render worker, a fixed cost of 2.0 s per request) every arc-on command of 50 ms or longer stayed within it, 133× at 500 ms to 177× at 50 ms. The fixed cost per request was 6.73 s with one camera and 12.07 s with three, an observation alone 1.05 s and 5.50 s, and the fitted physics with the arc burning 94-131 wall seconds per simulated second: other jobs on the GPU slow the physics as well as the renders.

A policy that pauses. The render worker is kept 600 s after each request (`renderer_options.idle_release_s`; it holds about 4.2 GiB of GPU memory meanwhile, and a request after a longer pause starts a new one, 10-13 s more). A worker released 10 s after a request would make a 50 ms command after a 15 s pause cost 303-495×. E16 also sends commands after 15 s pauses: 100 ms commands 191-212× with one camera and 249-269× with three, 50 ms 212-226× and 341-346×, without images 149×; the back-to-back costs. The policy's runtime note (`capabilities.hardware.wall_time`) is built from this file and quotes these numbers with the load it ran under (a contract test checks the derivation).

The runtime target is 200× wall time per simulated time. The simulations without the API are within it: at most 90× for a whole robot weld without rendering, 114× for E12's pool-every-2 ms case, 118× for the slowest stress case, and the films at 51-186× (the nominal four-view film 185.7×). **Not within it:** arc-on commands of 100 ms through the public API with images, under the load above (one camera up to 220×, three up to 272×); 200 ms commands with three cameras (191× median) and every command of 500 ms were within it.

Every number except E16's was measured while the GPU was shared with four to six other jobs (E16 ran with the GPU 100 % busy with another environment's render workers), the CPU quota of 16 cores at a load of 9-18. The thermal solver dominates the physics: its explicit substeps at the 0.25 mm cells' stability limit run on the GPU inside every 1 ms tick while metal is hot.

17. Interfaces

One registry path (`gym_anything_robotics/registry.py`: `welding`, aliases `weld`, `mig-welding`, `gmaw`, `weld-steel-joint`), so `gar.create`, `gar serve` and `gar run` construct the same backend (`gym_anything_robotics/backends/newton_welding.py`). Discovery: `gar list`, `gar inspect welding` (the deployments, from `robots/catalog.py`; a robot that cannot do a task is refused with the reason when it is requested: the robot and task are checked together at launch, before any session record exists, by `welding.variants:check_launch` through the catalog's `launch_check` hook, and the launch check also runs every backend option through the backend's own checks, `newton_welding.check_options`: unknown options, policy cameras, renderer options, config fields, seed and variation, so `{"cameras": ["hero"]}` and `{"phantom_torch": true}` are refused before a session record too), `gar tasks welding` (the catalogue `welding/benchmark.json` with the resolver `welding.variants:task_resolve`), `gar doctor welding` (`welding/doctor.py`: packages, CUDA, the Isaac render interpreter, the robot assets of every deployment (UR10e, FR3; the Panda and the G1 as optional rows naming dual-panda and g1), the pinned visual assets, the tacked start-state cache; `python -m welding.doctor --robot ALIAS` judges one named deployment and lists every deployment's asset status with a hint).

The `tool_io` robot action (`gym_anything_robotics/actions.py` `ToolIO`, decoded strictly in `action_codec.py`, listed in `ACTION_HELP` and in the policy allowlist `policy_access.ROBOT_ACTIONS`). It carries a robot controller's outputs to the process tool an arm holds and then runs exact physics ticks with the joint commands held. The welding backend declares the outputs, their ranges and the arm in its capabilities, and rejects anything else before any side effect (`ActionRejectedError`): `arc_on`, `gas_on`, `touch_sense` (booleans), `wire_feed_m_min` 1.3-25, `voltage_v` 10-44, `wire_inch` forward | backward | stop. There is no `process` output: the mode follows from the operating point (§5), and `{"process": ...}` is refused as an unknown output. The `gripper` action is offered only by deployments with a Franka Hand (ur10e-fr3, dual-fr3, dual-panda). A gripper command on a hand-less deployment is refused as an unsupported command (HTTP 422, "Robot command is unsupported; no action executed. Consult the robot capabilities."), not with GAR's generic 403.

A catalogue switch releases the world it replaces. `WeldingEnv.close()` breaks the world's reference cycles (its Warp arrays are freed at once), the backend runs one garbage collection (0.5 s) for the cycles inside Newton, and the first import of Newton happens before any world exists (Newton keeps an import error's traceback, and with it the frames on the stack). Memory stays flat at 1.1 GB of GPU and about 2.4 GB of host memory over eight switches; without the release each switch would keep +256 MB of GPU and +395 MB of host memory (contract tests `WorldReleaseTest`, `WorldReleasePhysicsTest`).

The policy's proprioception adds the source's feedback as numbers under `tool.*` (arc on, current, voltage, actual wire feed, touch, fault, gas; booleans as 0/1).

What the policy key can do: read the instruction, the capabilities and the action help; observe the joint state, the tool feedback and RGB from `bench`, `side` and `weld_camera`; act with joint targets, Cartesian pose or delta for either arm (for the torch arm the frame is the torch work point), the Franka Hand (position or force mode) and `tool_io`. What it cannot do: read the world state, the evaluation, the bond, the cells or the diagnostic cameras, take checkpoints, or apply any force; the assisted fixtures (the tack fixture's phantom torch, E7's phantom pull) are not reachable through the backend.

Contract tests (no GPU: catalogue and resolver, registration and aliases, the six deployments listed in order and their rigs as ids, setup objects and paths, launch-time refusals, the Panda profiles and bare-arm URDF, the torches, the G1's stance within its URDF limits, arms and hands per deployment, the policy's runtime note derived from E16's result file, every invalid configuration refused before the world is built, `tool_io` and motion rejections executing nothing, the restricted policy surface; with `GAR_WELDING_PHYSICS_TESTS=1` a live position-torch session, a gas-valve reset check of every task and the rigid cells built): 83 tests with 342 subtests, 0 failures, 0 skipped. They also cover the overhanging surface against the weightless pinned drop and read as overlap, no weld metal without a face on steel through a back-step strike, every backend option checked at launch, the replaced world released across catalogue switches, the runtime note quoting the paused policy and the GPU load, the G1 refused the tack task, the draft's shielding law against Kobelco's chart and the wire balance.

Evaluation on demand and timeouts. Owner states and command results do not carry the evaluation (`evaluation: null`, `reward: null`); `GET /v1/evaluation` and `gar verify` evaluate the current tick on demand, and the backend option `state_evaluation: true` restores an evaluated state. The state is read once per tick. A single legal request can outlast the 120 s HTTP clients often default to (an arc-on command of seconds at 40-60 wall s per simulated s, the first build of the tacked weld state): the backend declares `client_timeout_s: 3600`, which the `gar` CLI uses unless `--timeout` is given.

Checked through the delivered entry points. E16 (FAIL on the runtime limit, §16) starts `gar run welding --task fillet-2f --policy-key` once per camera set (`bench` alone, then `bench`, `side` and `weld_camera`). An HTTP client with the policy key sends two observation-only requests, the source's schedule, an approach, arc on and the strike, then 17 arc-on Cartesian travel commands along the seam of 500, 200, 100, 50, 20 and 10 ms (each

returning an observation with the set's images), three 1-tick `tool_io` commands alone and three batched with a 50 ms motion, and arc off. Per set: one strike, 244 A mean (288 A at most), no gun contact, no wire touch, no fault, no protective stop. Every command did what it should; it is the wall time per command under the GPU's load that fails (§16). The session dashboard (8 of 8 checks) is driven in a browser with the policy key: connecting does not step the world; the schedule, a 20 mm torch move and 0.2 s of wire inching each run their exact ticks and reach the source (the stick-out grew from 11.5 to 19.8 mm); an out-of-range `tool_io` (80 V) is refused with HTTP 400 `ActionRejectedError` without stepping; no browser script errors. The screenshots show the bench camera's view of the clamped coupon and the gun's nozzle, the task text and the committed or refused commands.

Operator page (`welding/manual.py`, `manual_static/`). A keyboard page attaches to a running session with the policy key (robot cameras, the source's feedback, jogs of the torch work point and the hand, the hand's position and force modes, the source's set points, gas, touch sensing, wire inching, arc on/off, joint and Cartesian targets) or the owner key (adds the evaluator, the privileged state, checkpoints, the diagnostic render and a three.js view of the colliders posed from `/v1/state` and `/v1/geometry`). Every key or button is one policy action through GAR; nothing on the page moves the world except through the robot or the source's outputs. Its verification (headless Chromium against a live session) measures each command's effect in the owner API's history (jogs to 0.01 mm, the hand's opening, the wire's stick-out, refusals that leave the tick unchanged, the policy boundary, a checkpoint round trip, no key reaching the browser): 33 of 33 checks pass, and 33 of 33 on the dual-Panda deployment. The G1 has no TCP readout or 3D view on the page (a floating base: its torch pose is not a function of its joint positions alone); the page says so.

Per-tick physics trace (`welding/trace.py`). Every action through GAR records one sample per 1 ms tick (joints with targets and servo set points, the torch work point, the source's state, net contact force per arm and gun contact pairs, the bond, the pool's volumes and the energy ledger) plus events (tool I/O, strikes, touches, circuit, faults, protective stops, bond made or broken, gun contacts); `GET /v1/history/{id}/trace` and `/trace/events` (owner key only; the policy key gets 403) and `gar export --include-traces`. Measured cost: +16 % cold, +1.6 % with the arc burning, +3.2 % warm; the trace's per-tick ledger agrees with the process's own.

Variations (§13): `gar run welding --variation N`, `--select split=test --select selection_seed=K`, `/v1/variations`; E22 builds a pre-registered sample of nine entries through the backend and checks the start state against the entry and a hands-off rest.

Motion guard. Cartesian commands honour `safety.stop_on_arm_arm_contact` (default true): with two arms the command runs tick by tick and ends at the first tick the arms touch, both arms holding; the step's info carries `guard_stop`. The other guards (`stop_on_contact`, `stop_on_robot_contact`, `stop_on_grip_loss`) are refused before any tick. Approaching the gun with the FR3's hand at 60 mm/s, the guard ends the command on the contact's first tick (85 N), the tick on which the FR3's collision reflex also fires; at 1 m/s the first tick carries 2.3 kN. In this cell the arms' protective stops already act at first contact; the guard adds the early end of the command and its report.

18. Limitations and work not built

Not built (open work, not "accepted limitations"):

- **Presets B and C** of the design (copper brazing with an oxy-acetylene torch and a nitrogen leak test; electrical soldering), and **steel-cell tasks A3-A6** (pipe on a positioner, vertical-up, sheet seam, stick electrode). The parameters for B and C are recorded as reference entries in the ledger and read by nothing.
- **Transverse weld shrinkage.** The angular distortion is built (§8, E25); the joint's transverse shrinkage, about 0.8 mm per fillet (`SHRINK_TRANSVERSE`, a reference entry), is not. No law for two short tacks was found: the volume scaling gives them about 0.1°, a scaling by weld size alone about 0.5°.
- **A run-on start.** The grid is 0.5 mm along the weld only from x = −2.5 to 152.5 mm; beyond, its cells grow to 3-9 mm (`process.build_grid`), so an arc started off the joint's end (the trade's run-on tab) lands on cells far coarser than the pool model was checked on. No run-on tab exists in the cell either.
- **A G1 tack.** The G1 is not offered the tack task (§9): no grip strategy for the Dex3 (its thumb hangs 42 mm below the palm), whole-body path or balance controller that holds the bracket's top edge while the torch reaches the tack has been built.
- **A single-arm clamped tack**: a fixture clamp holding the bracket so one torch arm can tack; not built.
- **The operator page's TCP readout and 3D view for the G1** (the page works for the G1 through jogs and joint targets; its kinematics cover the fixed-base arms, the dual-Panda included). The catalogue's reach check covers all six rigs; the G1's left hand cannot reach the bracket's top edge from its standing waist.
- **The hose package's load on the gun** (drawn, not simulated). No sourced mass or stiffness per metre of a cobot hose package was found; the order of magnitude (about 2 N and 0.5-1 N m at the gun with a support 0.5 m away; derived from a 1/0 AWG conductor's copper) is within the arms' servo rejection.
- **A stray arc's metal and heat on the body it burns on.** A stray arc (§11.3) books the wire it melts and its heat (`stray_arc_m3`, `stray_arc_j`); the metal is given no shape or collider on that body (a 2 s stray arc leaves about 235 mm³, a hemisphere about 5 mm across, which would shorten the arc as it grows) and the body no temperature. Stray arcs are not drawn in the films.
- **Welding a loose bracket away from its fit-up place.** While an unbonded bracket is away from its fit-up pose its cells leave the thermal grid (§8, E24): nothing welds to metal that is not there, and an arc on the moved bracket is a stray arc (booked, not modelled thermally); it can be welded again once it is set back within 0.8 mm.
- **A bond whose strength depends on the load's direction.** The bond's transverse strength, 0.852 F_EXX, is one value (§8); Thomas & Tousignant's single-sided fillets broke at 0.37-0.47 F_EXX when the load bent the root open and at 0.96-1.05 F_EXX when it bent it closed, so the bond over-predicts peel by 1.8-2.3 times and under-predicts root-closing loads by 11-19 %.
- **A near-infrared weld camera.** The policy's weld camera is a visible-light HDR camera, in which the pool is a mirror (§2); published process-camera frames with a bright, temperature-lit pool are near-infrared monochrome. A NIR camera would need the arc's NIR line emission and the camera's filter from sources (liquid iron's 807 nm emissivity, 0.394, is already in `render.py`).

Model limits (stated, sized where measured):

- **The bead on plate humps** (E3, §11.8). On the flat plate the pool lays the reference's bead as alternating flat slabs (8-12 mm² of deposit, 1.2 mm high) and domes (30-46 mm², 4-5.5 mm high) every 3-5 mm along it; the metal is exact (939.5 against 938.5 mm³), its placement is not, and E3's median section holds 13.0 mm² against the reference's 23.4. The cause is not established. With the overhang's shares in proportion to what each slice holds (instead of each slice first taking what it held past its face last update) the bead's first 14 mm is even (20-29 mm²), but it thins to 9-17 mm² and humps again over its last 10 mm: the share rule contributes and is not the whole cause. 801 of the bead's 1221 overhang updates ended unconverged. The fillets on the tee do not hump (the nominal robot fillet's size 6.02-6.70 mm over x = 20-120 mm), and no task welds on a flat plate; E3 is the check that shows it. Not fixed.
- **Single cells of weld metal can be left in the air** (found by `sections.floating_weld`). The cells are set from each slice's surface by their centres; where an overhanging surface narrows below a cell's size, a cell under it can be left with no metal beside it. Usually the next update takes it back; if the slice freezes first it stays: 1-5 one- or two-cell pieces (0.06-0.16 mm³) in 3 of 27 robot demonstrations (`dual_fr3`, `aim_up_2mm`, `fast_travel`), 3 cells (0.09 mm³) in E20's touch-sensing case, one inside a groove in the bracket, and 15 pieces (1.3 mm³) at the edge of the pool's region in E20's 15 s park. The weld cells are about 1000 mm³ in a fillet. The cause-level fix, laying only cells that connect, in the section, to metal, is not made. An overhang is not cut at a fixed radius: the pool owns all liquid connected to its liquid (§6).
- **Cold starts: what remains.** The pool's surface overhangs its toes as one bulge pinned where it touches, and a later pass remelts what its pool reaches (§6). The robots' tacks (the back-step start) are fused to within 0.5 mm of their run's first section and carry 0.5-2.0 mm of overlap (the largest a lip at one strike point), and fail D1.1's tack checks on one to three sections over the convexity limit and, on the two-arm rigs, 6.45-6.5 mm against the ¼ in tack size (§14). E26: no starting technique passes every tack check; the back-step starts come within 0.75 mm of overlap. Fillets: the nominal programs leave 1.25-1.5 mm of overlap over the

first tack's start, which a program that aims 1 mm onto the plate there washes in (`aim_tacks` passes, §14). Sized against reality: not done (no measured geometry of a spray cold start on 10 mm plate was found). Every saved fillet's own start carries more than one cell of it too (0.5-12.25 mm over its first 5 mm, 20 of 20, with convexity over Fig 5.4 on some of the same sections). The fillet evaluator therefore accepts a start's overlap within 5 mm of an end, outside every crater, up to two cells (0.5 mm): what the two starts shown to clear the tacks leave there (`aim_tacks`, `travel_5p0_x0_1`). Every other start is rejected, and so is a start's convexity (§13). That leniency follows from this unsized start geometry; what lies there is reported in every evaluation (`diagnostics.start_zone`). A weld started inside the span is judged at one cell.

- **A pool larger than its region.** The pool model changes cells within 14 mm along each member. A spray arc parked for seconds outgrows it: in E20's 15 s park (§15) up to 561 mm³ waited unplaced, the frozen metal ended 2.4 % over the deposit and 15 cells' pieces stood at the region's edge; the arc stubbed after 9 s when the bead reached the wire. Parked arcs stub (E8 (d) and (e) fail their predictions, §15); a pool standing tall under a parked wire is what a real parked arc builds, but the outcome was not compared with a measured one.
- **At Lincoln's published ¼ in spray procedure the fillet is 2-3 % undersize.** The robots lay a median size of 6.22-6.25 mm (≥ 6.35 mm on 18-23 % of the length; §14), with the metal exactly the deposit. The arc's heat spread σ = 4 mm is an open parameter (E3 reports what it moves) and was not tuned to close this. At 4.8-5.2 mm/s the size reaches 6.35 mm on 97-100 % of the length.
- **The crater is not filled to the specified size by the programs' 0.5 s dwell.** D1.1 asks for the crater filled to the specified size (6.35 mm). The model's last metal to freeze is short by the liquid's 3 % shrinkage over the last pool (§6), and a 0.5 s crater dwell leaves 5.67-6.25 mm over the last 10 mm; a 1.2 s dwell fills it (6.45 mm in `crater_1p2`, 6.73 mm in `aim_tacks`). A longer crater dwell is a technique the task leaves to the policy. The bead's own last 1 mm is left out of the crater at a bead end, and the crater is the 10 mm centred on where the pool froze last (§13). On the UR10e and the dual FR3, at 29 V aimed at the root, the 0.6 s dwell's fall-off then lies in that last 1 mm, and those runs pass the crater check at 4.8-5.93 mm/s (judged from the saved sections). The dual Panda's (6.00 mm), the G1's at 5.93 mm/s (6.20 mm) and the UR10e's at 31 V or aimed 0.5 mm up fail.
- The pool is 2-D slices with longitudinal sharing: no Marangoni-driven shape, humping, droplet-by-droplet transfer, arc flicker or surface ripples; penetration comes from the enhanced liquid conductivity and the droplet heat source.
- **Mass is conserved within 0.25 % in the procedure window and 2.4 % past the pool's region.** The robots' fillets end within 0.02 % of their deposit and E26's tacks within 0.24 % (frozen metal keeps the expansion it froze with where no later liquid takes it back, §6); a 5 s parked arc +0.5 % (process only); the source's maximum parked until it stubbed +0.29 % (E8 (e)); E20's 15 s park +2.4 %.
- **Collision shapes are convex approximations of the real parts.** The nozzle is a 32-sided cone circumscribing the real one (at most 0.05 mm proud); the gun body and mount are cylinders and the neck a capsule (authored geometry: no drawing was available; masses and the TCP are read); the clamp pads are cylinders that MuJoCo gives one contact point against the plate (adequate for a pad on a slide). A nozzle pressed into the parts penetrates 0.04-0.07 mm at 67 N.
- **A pinned arm is freed by backing it off.** A protective stop refuses the arm's commands for 1 s and then unlocks it, standing for the robot program's own unlock (UR's dashboard command "unlock protective stop", allowed at once except 5 s after too-frequent stops; libfranka's `automaticErrorRecovery()`: both read). For 2 s after the unlock, the load that caused the stop may persist (up to 10 % above it) without a new stop, so a command can move the arm away; pushing further stops it again, and an arm left pressing is stopped again when the window ends. Without the window, an arm stopped while pressing into an obstacle above its threshold would re-trip on the next step after every unlock. The window and margin are authored: how the real controllers treat a load that persists after the unlock is not in the sources read.
- **Weld metal collides as boxes** (§6): one per 2 mm of joint, refitted when its cells change and its slices are solid; 0.9-1.8 mm of empty space above the metal mid-bead, 3.7 mm at its start and up to 6.5 mm over its last 6 mm (§11.11); a liquid pool keeps its segment's last solid box. Beads collide with robots, the gun, the hand and the clamps, not with the bracket or plate under them. A refit may not reach into a body deeper than the box in use did: under a resting body the box's outer face is pulled in, or the box in use is kept until the body leaves, so there the collider can lag the metal by up to a cell plus the box's own coarseness.
- **The standing G1's residual jitter and its open causes.** Newton's float32 copy of MuJoCo's state leaves 0.02-0.7 µm/s of positional jitter with no direction, a few µm of span. The gravity feed-forward is computed every tick because, held 5 ms, it drove a sustained vibration (pelvis 85 µm peak to peak); the mechanism is consistent with a sampled feedback with delay but not fully established: a feed-forward computed from the commanded configuration alone, as Unitree computes it, made the robot slide at 6-8 mm/s in one implementation, for reasons not found. The G1's contact gap is 0, as every other shape's; at a 0.1 m gap MuJoCo's box-box routine returned one contact with its depth's sign flipped, and whether it is safe at gap 0 in every configuration was tested only in one (±2 mm).
- **No arc-length limit of its own.** A long arc is bounded only by the contact tip: at 44 V and a 30 mm contact tip to work distance and 12.7 m/min the model sustains a 26 mm arc at 357 A (`arc.CVSource.steady`). No

source was found for the length at which a real CV arc goes out, so none is applied (§5).

- **The clamped plate slides at 951 N, 13 % below what its contacts should hold** (a push test's measurement). Measured here: each clamp pad presses 385.4 N (its spring's equilibrium, 3.6 % under the 400 N preload), MuJoCo uses μ 0.8 at the pad contacts and 0.6 at the table (read from the solver's contacts), so a push through the plate's centre should need 0.8 × 770.8 + 0.6 × 800.8 = 1097 N. Where that push acted (an off-centre push lets the plate turn about one pad) was not reproduced; the remaining difference is not explained. Far above anything the UR10e (150 N stop) or the FR3 can apply.
- **The base plate creeps at about 4 nm/s at rest** (0.24 μm per minute, linear), 0.7 μm over the longest task's 180 s. Cause not determined (MuJoCo's soft contacts under the clamp preload are the candidate).
- The deposited filler's mass (≈ 25 g on 3 kg) is not added to the bodies.
- **The wire does not bend or sag**: pushed against metal it slides back into the contact tip, and free inching stops at 100 mm. The power source's case is not in the welding circuit (its output is insulated from it); a table bonded to protective earth, which would put the case in a loop, is not modelled.
- Section measurements are quantised to the 0.25 mm cells.
- **The bench camera sees the FR3 torch arm's wrist.** It looks straight down on the joint (§2). In the FR3 and Panda torch deployments the arm's wrist covers about the lower third of its view while welding, and the joint and the arc stay in view. A top view does not show the bracket's face or the fillet's legs; the side and weld cameras do.
- **The weld camera's pool looks folded, and two thin bright lines cross its frame** (Figure 2.2). The pool's near surface reads as crumpled rather than smooth, and two thin vertical lines of light run down the right of the image; their source in the renderer was not found.
- **The weld camera is a viewpoint without a body.** It rides on the gun 110 mm behind the work point and 38 mm above the plate, where the arc, pool and fresh bead are in its line of sight; a real process camera's housing (about 60 mm across) would not fit there and would sit further back with a longer lens. It has no collision shape and is not drawn.

19. Reproduction

From the release's repository root, after installing it as its README describes. `gar doctor welding` checks what the environment needs beyond Newton, Warp and MuJoCo (a CUDA device for the thermal solver, the Isaac render interpreter, the robot assets of each deployment, the pinned visual assets) and says how to provide what is missing:

```
gar doctor welding                                  # requirements of the default deployment, ur10e-fr3
python -m welding.doctor --robot dual-panda         # the same checks for a named deployment
gar tasks welding                                   # the tasks, presets and variation selectors
gar tasks resolve welding --selector-json '{"task": "fillet-2f"}' --output welding.task.json
gar run --task-file welding.task.json --port 8080 --policy-key   # a session (HTTP, dashboard) for a policy
python -m welding.manual --session <id>             # the keyboard operator page on that session
gar verify welding                                  # the task evaluation of the running session
```

`gar run welding --robot SETUP` picks a deployment, and `--variation N` or `--select split=test` a catalogue entry (§13). The tacked weld state is built by its fixture once per physics version (about 3 minutes on a quiet GPU, up to 45 on a busy one) and cached by a hash of the physics source (under the workspace's `.gar/welding-cache/`, else `~/.cache/gar/welding/`); every deployment and catalogue entry places it in seconds.